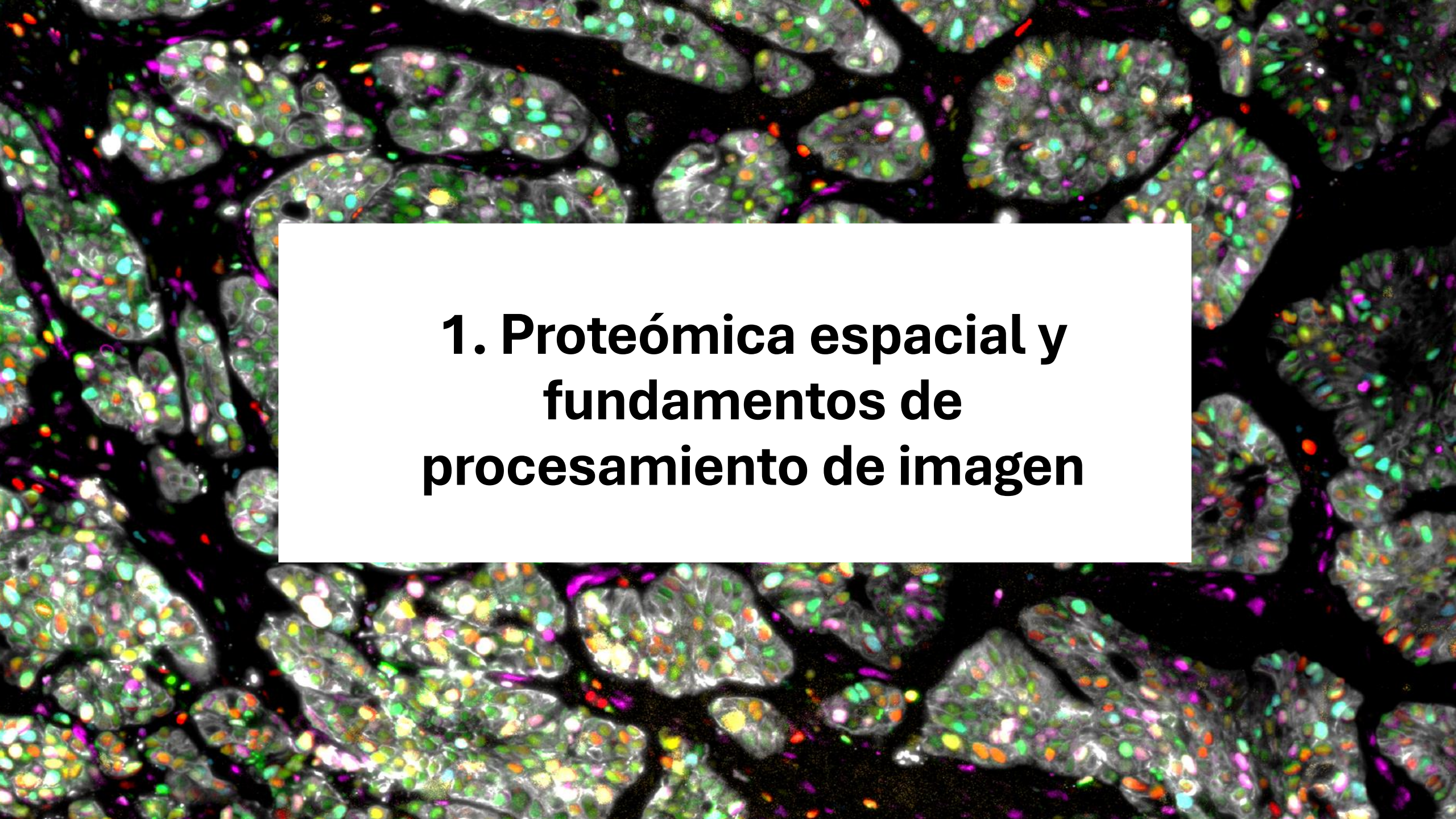


A fluorescence microscopy image showing a dense population of cells. The cells are stained with various fluorescent dyes, resulting in a multitude of small, bright spots in colors like green, red, blue, and yellow. The cells themselves appear as darker, irregular shapes against a black background.

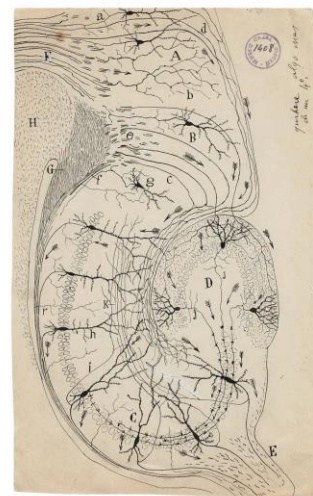
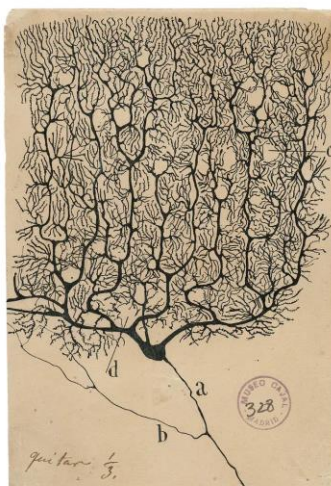
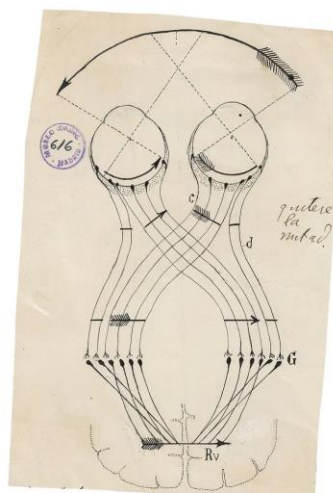
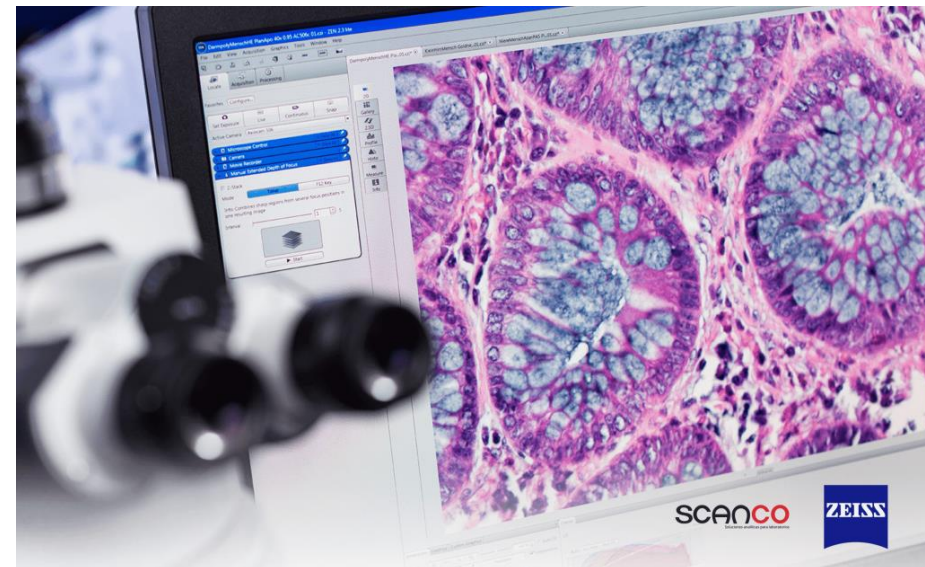
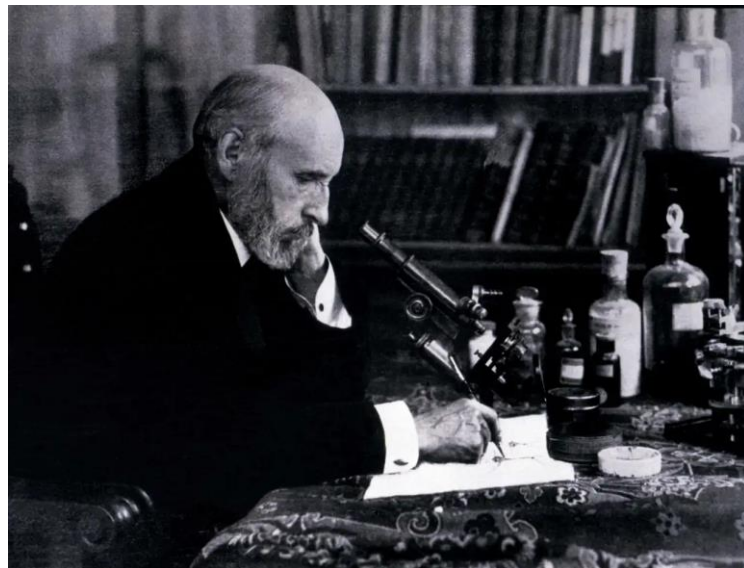
Fundamentos de análisis de imagen en microscopía

Ada Junquera, MSc
ada.junquera-mencia@helsinki.fi

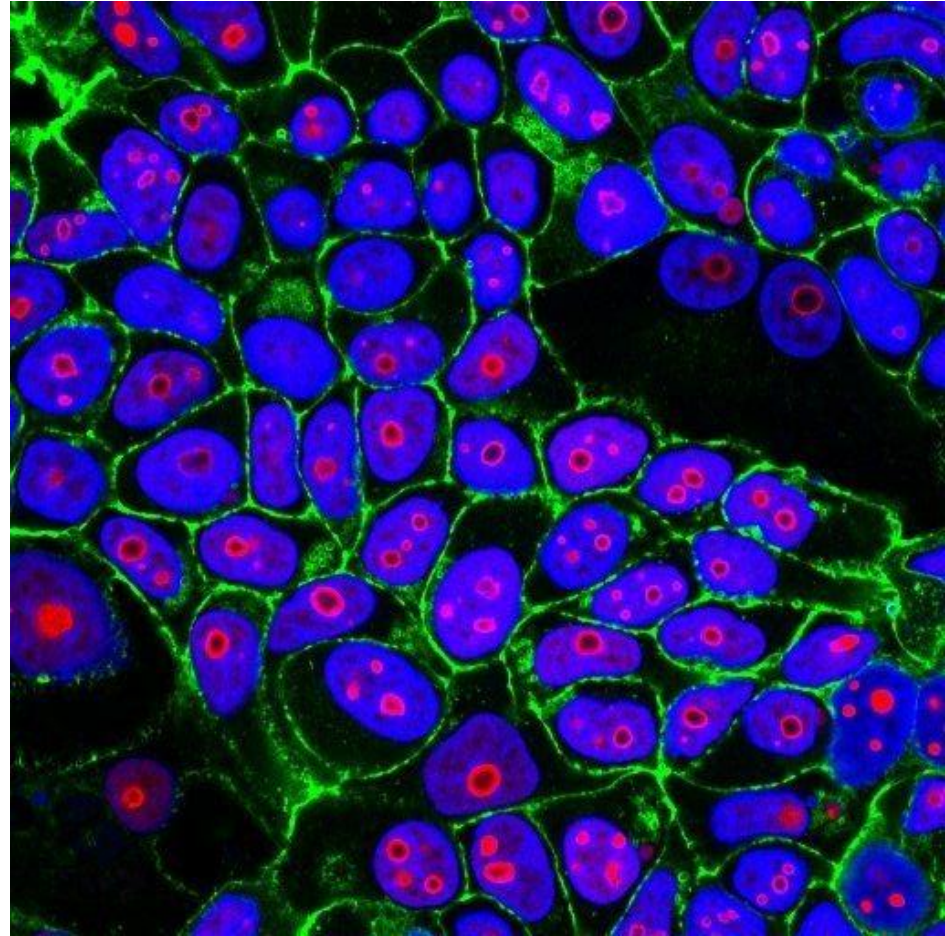
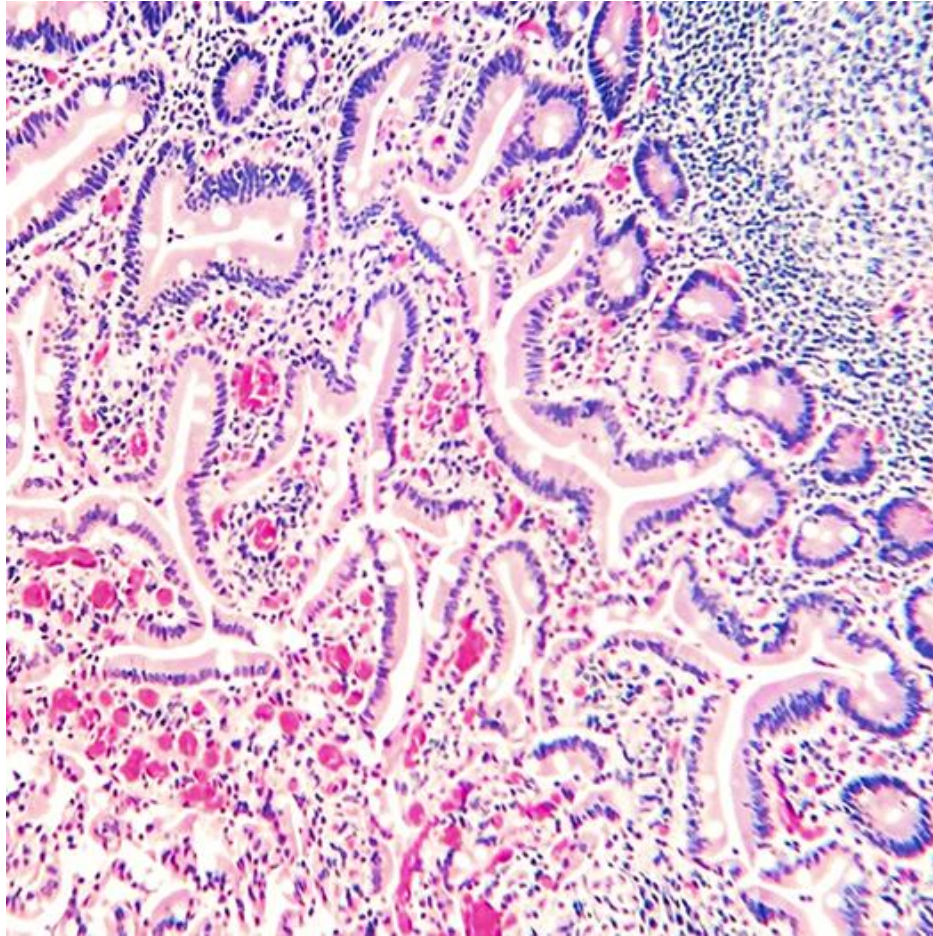
A fluorescence microscopy image showing a dense population of cells. Each cell is filled with numerous small, brightly colored spots in various colors including green, red, blue, yellow, and magenta. The cells are irregular in shape and are set against a dark background.

1. Proteómica espacial y fundamentos de procesamiento de imagen

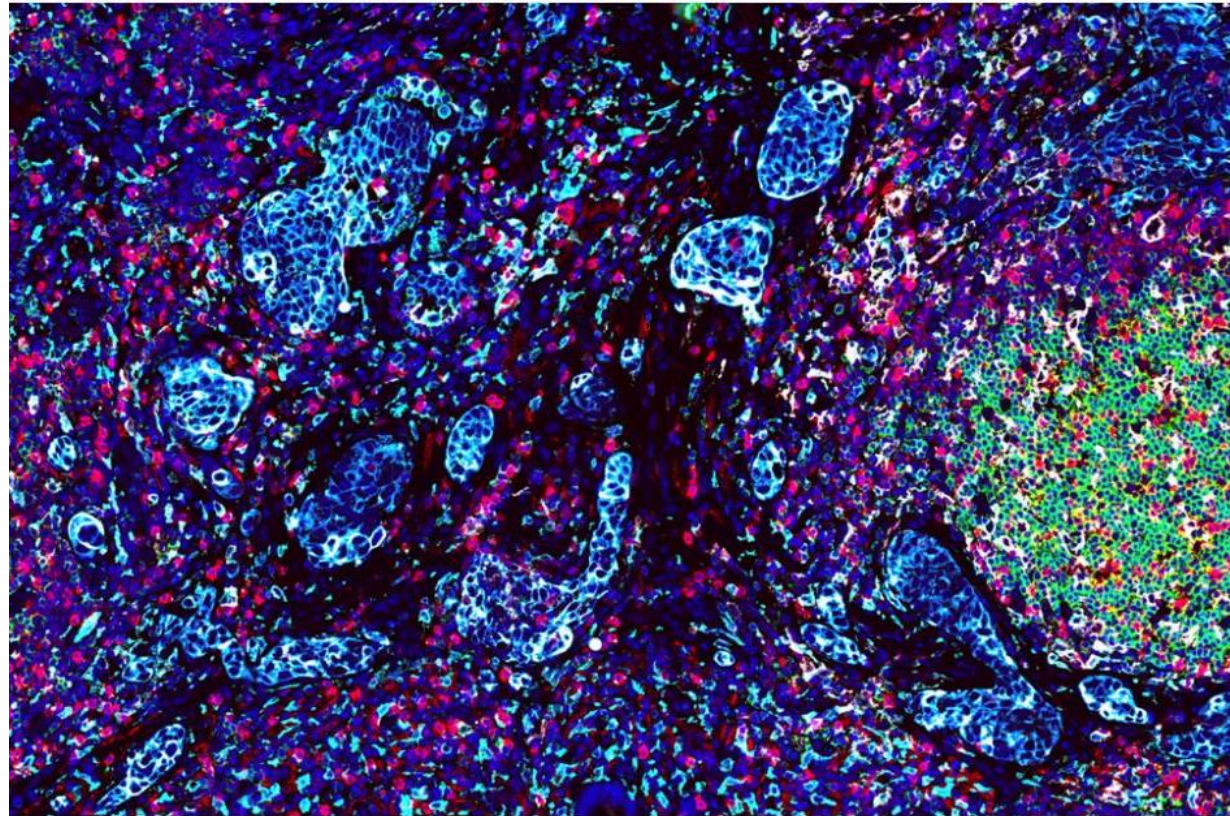
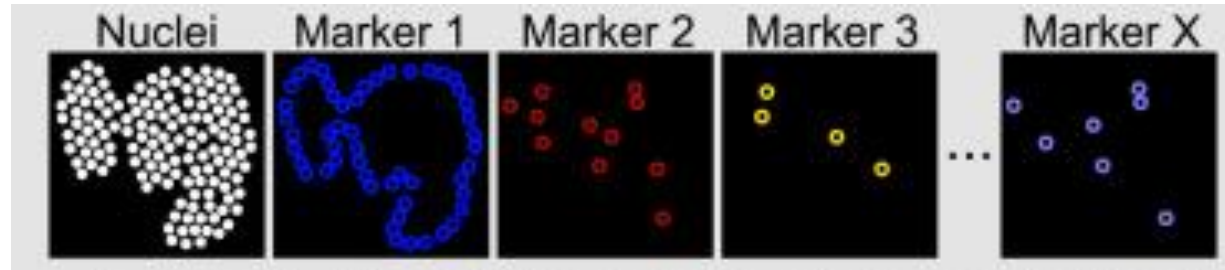
The history of microscopy



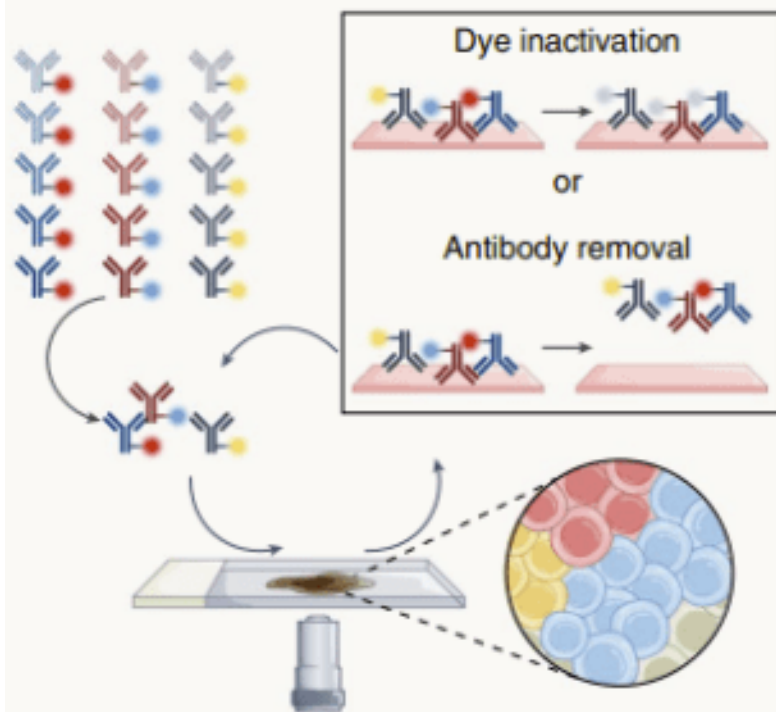
Limitations of Traditional Tissue Imaging



What is Multiplexed Imaging?



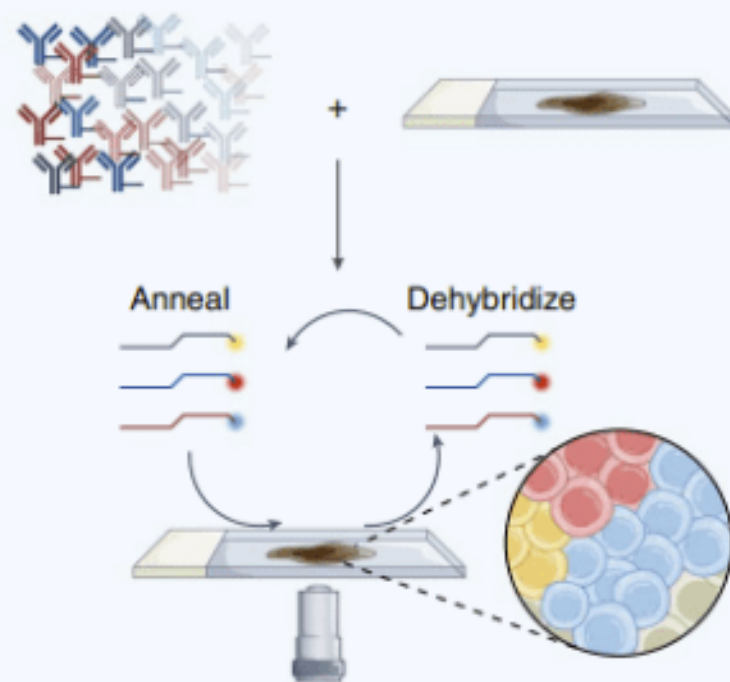
Fluorophore-labeled



Cyclic staining and cyclic imaging
Inactivation: CellDIVE, t-CyCIF, MELK, IBEX
Removal: MAP, 4i, SHIELD, SWITCH

All-in-one imaging
Opal IHC, hyperspectral, vibrational

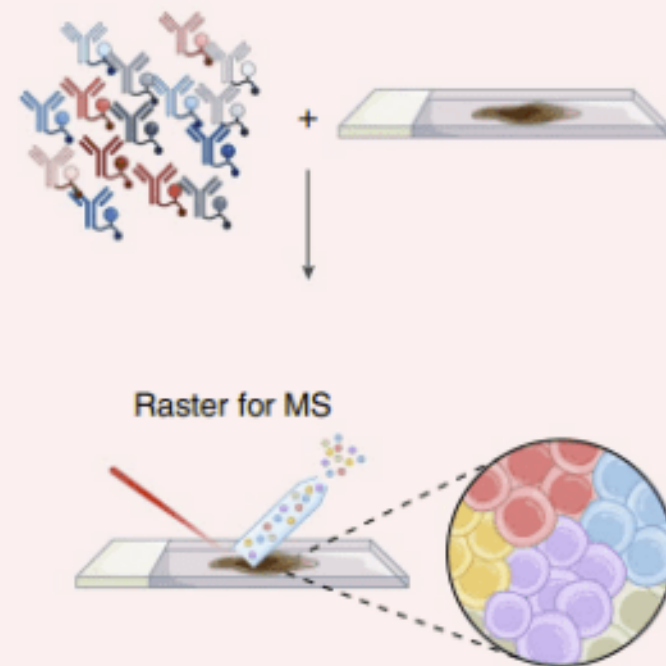
DNA-labeled



All-in-one staining and Cyclic imaging

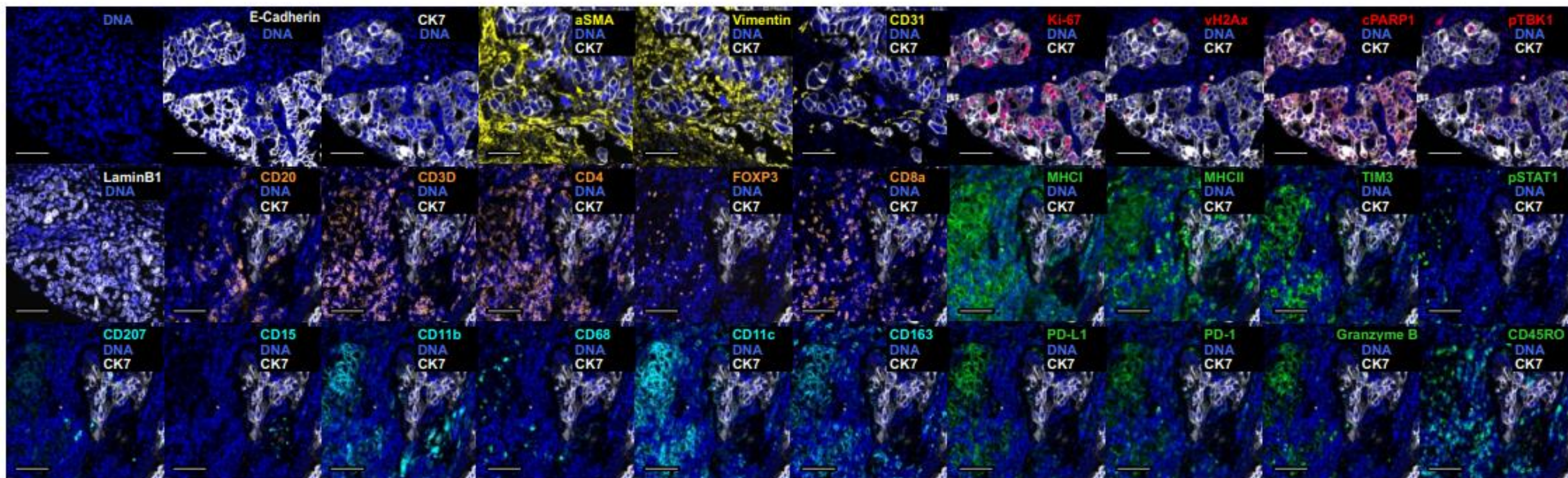
CODEX
Immuno-SABER
DEI

Metal-labeled

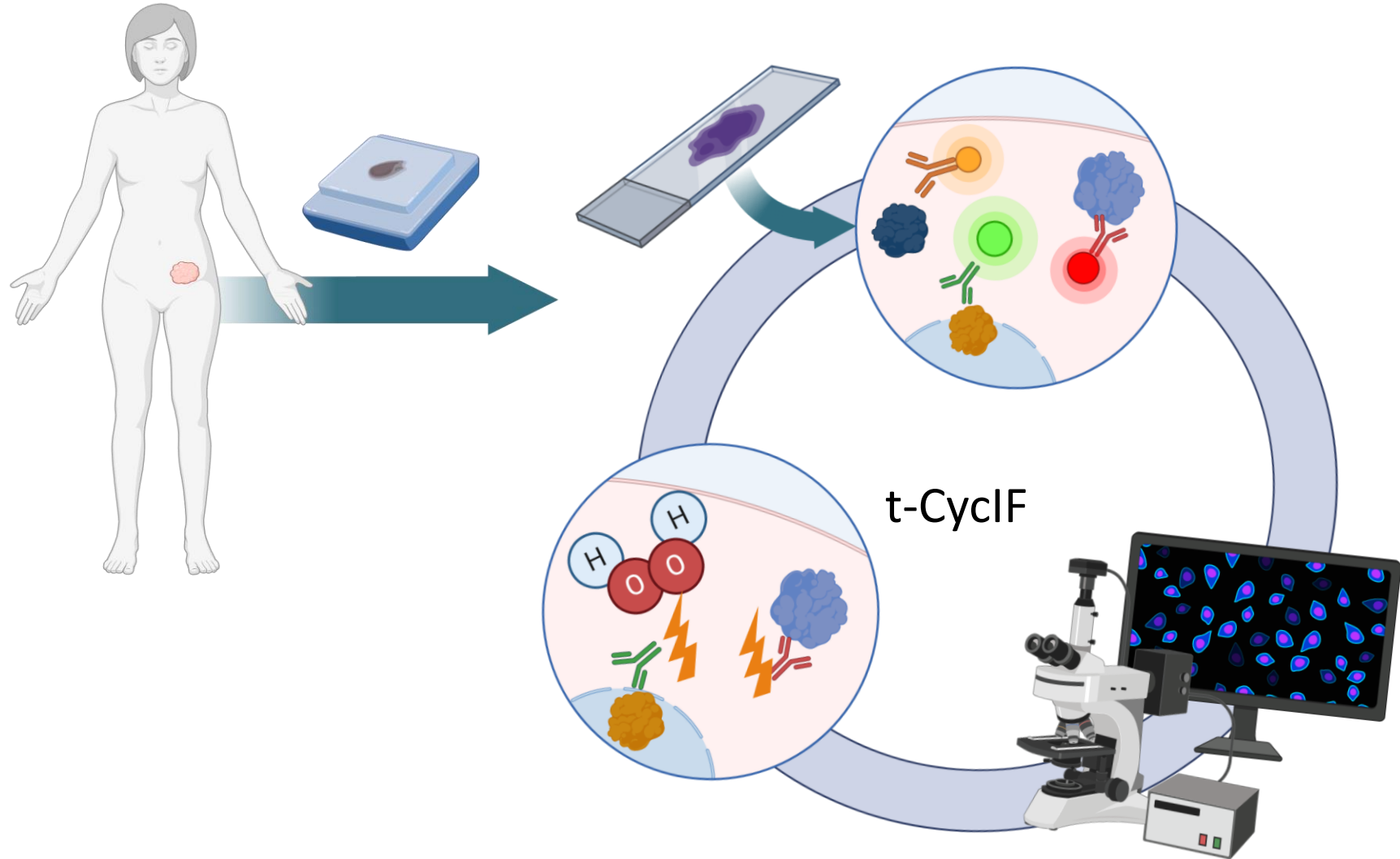


All-in-one staining and all-in-one imaging
IMC
MIBI-TOF

What is Multiplexed Imaging?

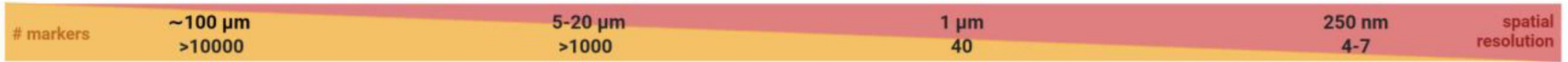


Tissue cyclic immunofluorescence (t-CyclIF)

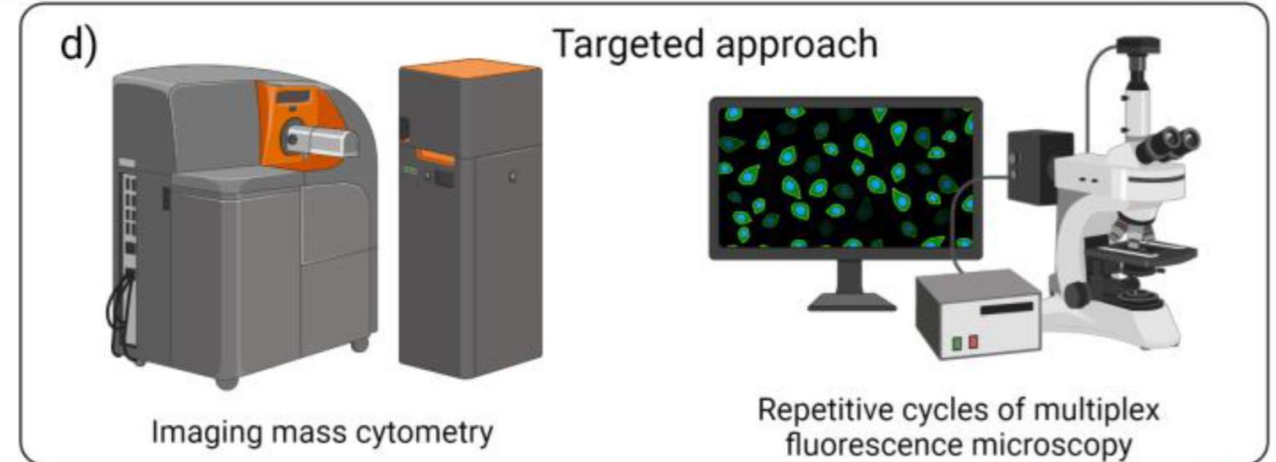
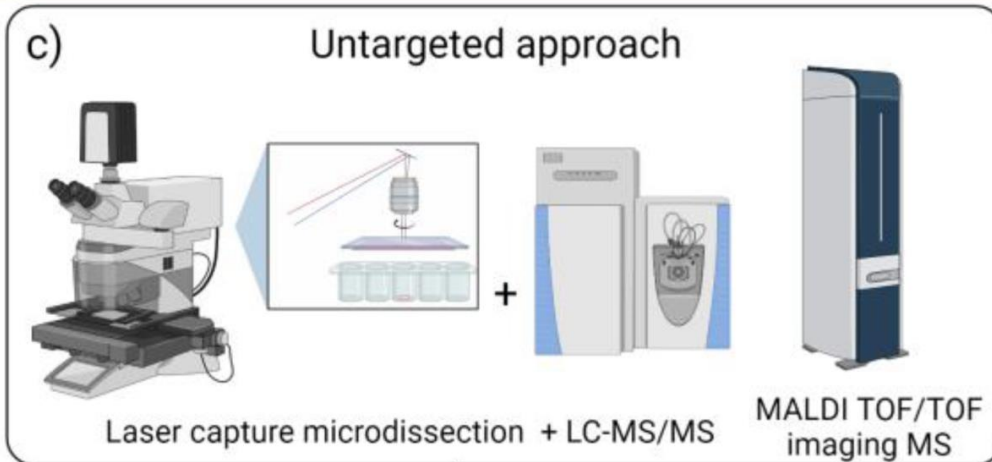


**Is highly-multiplexed imaging
a spatial proteomics method?**

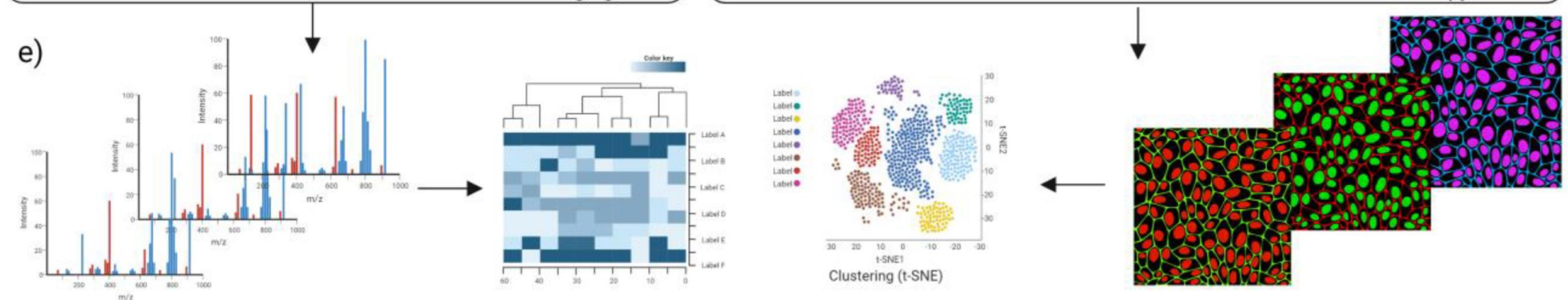
Is highly-multiplexed imaging spatial proteomics?



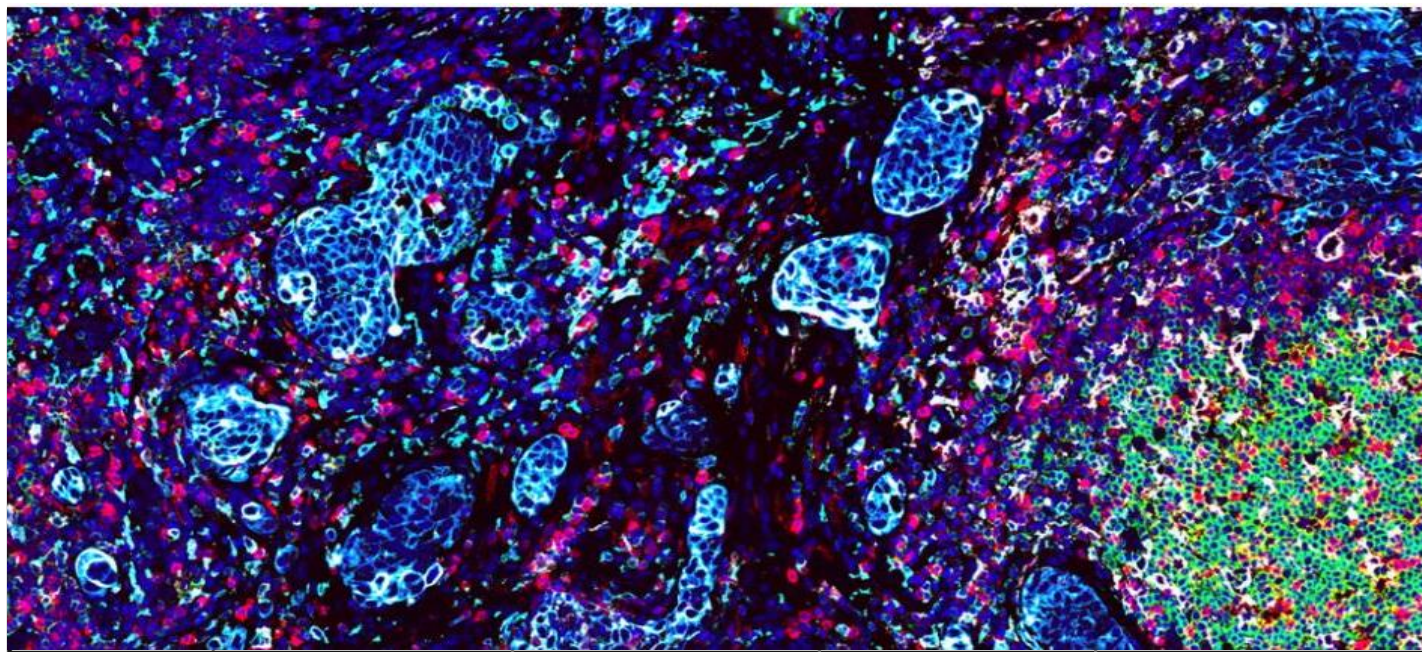
Spatial method



Data analysis



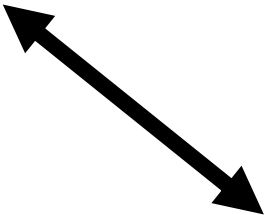
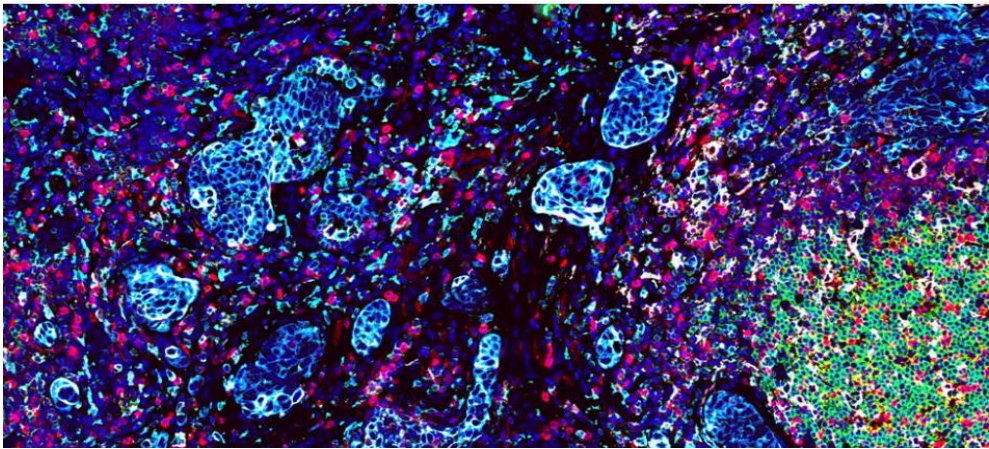
From Images to Insights



From Images to Insights

CellID	X position	Y position	Marker	Phenotype
1	48578.02	12079.58	291.29	Cancer
2	12508.93	18347.84	840.30	CD8
3	41332.14	16543.74	299.86	Stromal
4	34043.81	24965.79	444.17	DC
5	21639.99	21359.70	514.44	CD4

From Images to Insights



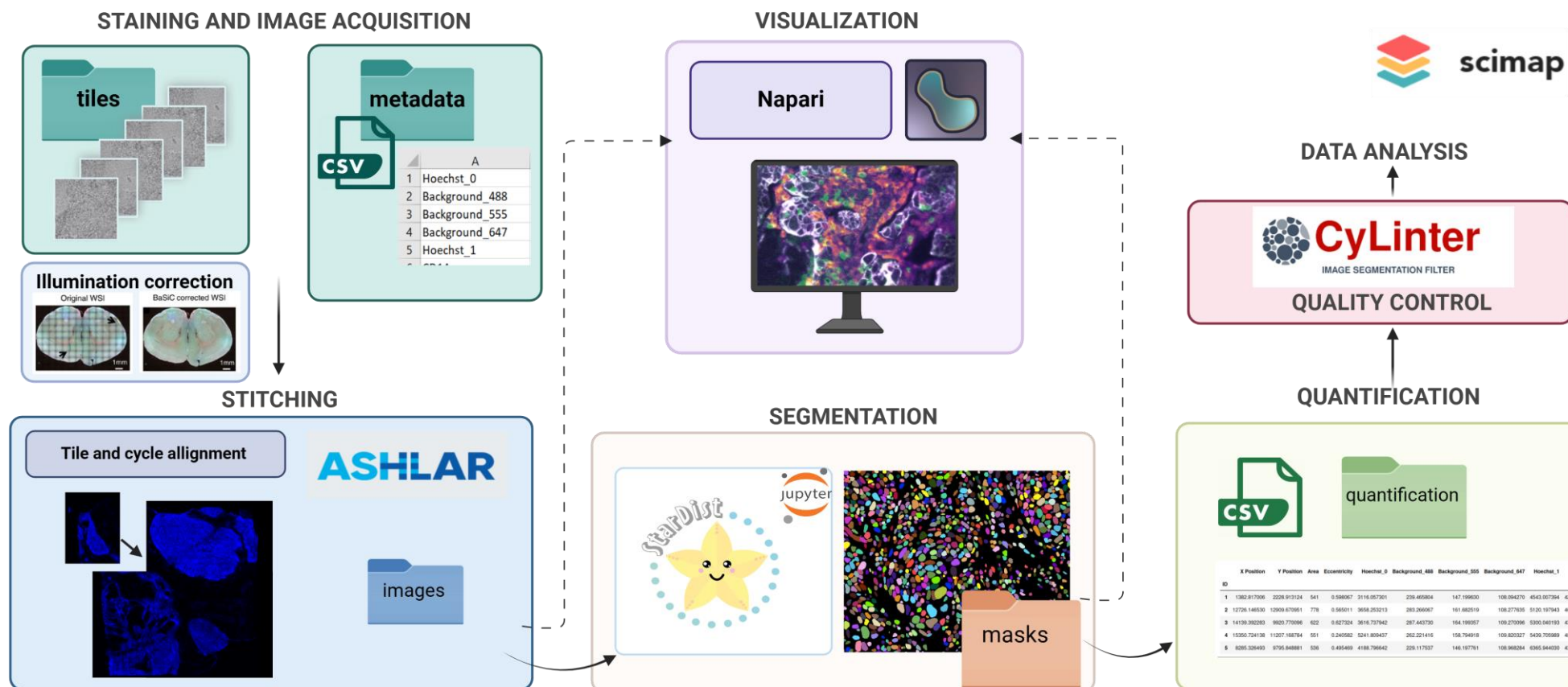
CellID	X position	Y position	Marker	Phenotype
1	48578.02	12079.58	291.29	Cancer
2	12508.93	18347.84	840.30	CD8
3	41332.14	16543.74	299.86	Stromal
4	34043.81	24965.79	444.17	DC
5	21639.99	21359.70	514.44	CD4



Data might look similar , but we must always use our eyes



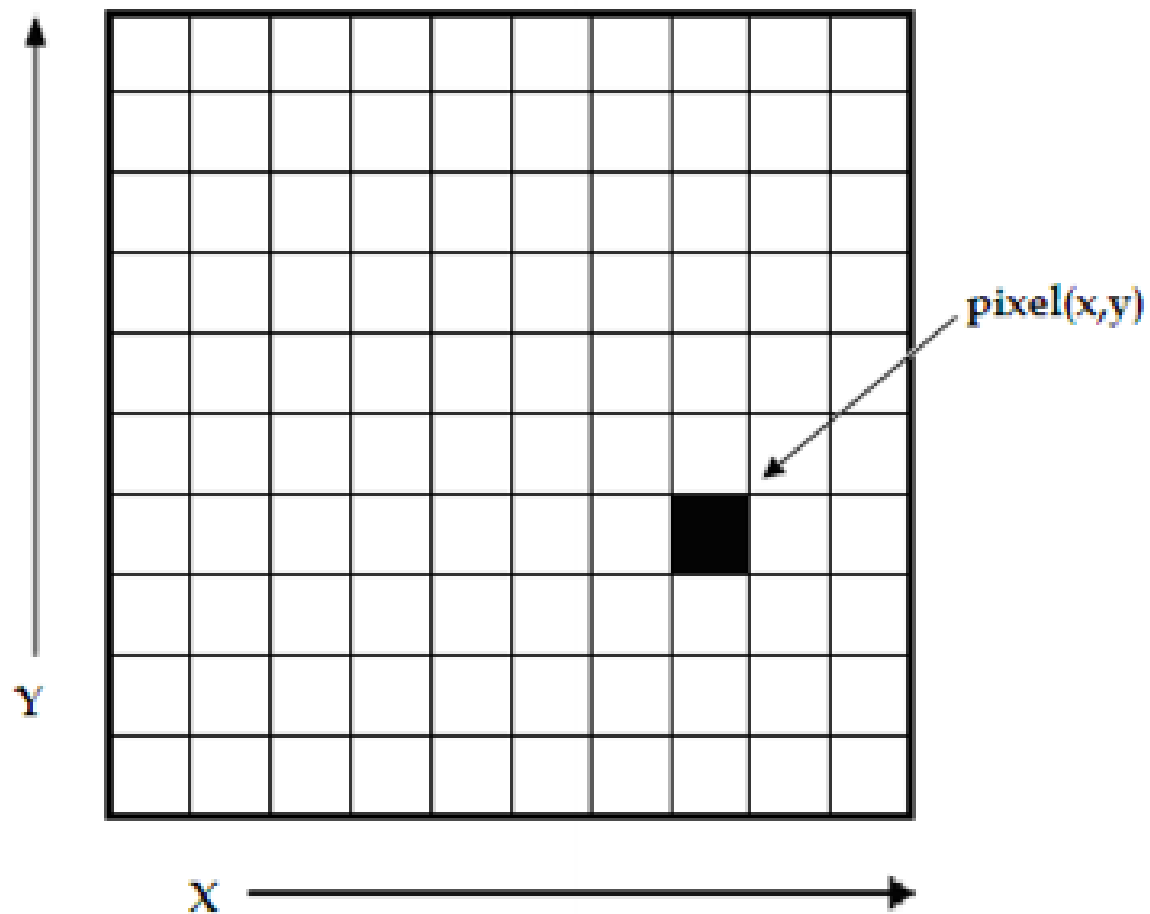
Image processing workflow



2. Computational image handling

What is an image?

Understanding images computationally



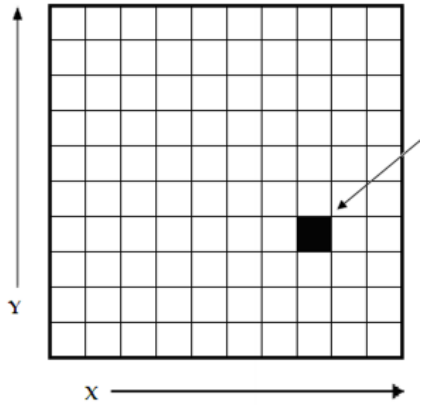
Grayscale Images



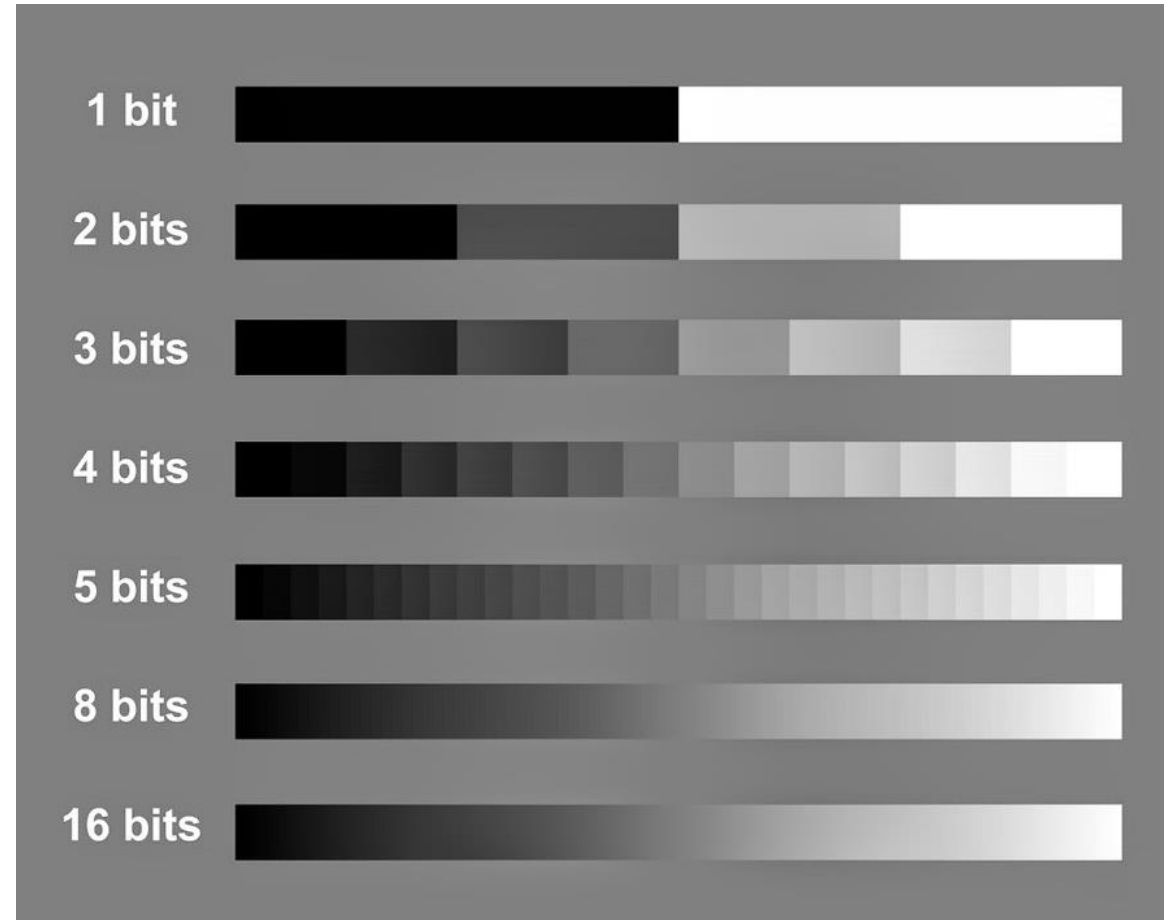
167	163	174	168	160	162	129	151	172	161	165	166
165	182	163	74	75	62	33	17	110	210	180	164
180	180	50	14	54	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	96	180
194	68	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	198	227	178	143	182	106	36	190
205	174	165	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	234
190	214	173	66	103	143	96	50	2	109	249	215
187	196	236	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

167	163	174	168	160	162	129	151	172	161	165	166
165	182	163	74	75	62	33	17	110	210	180	164
180	180	50	14	54	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	96	180
194	68	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	198	227	178	143	182	106	36	190
205	174	165	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	234
190	214	173	66	103	143	96	50	2	109	249	215
187	196	236	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

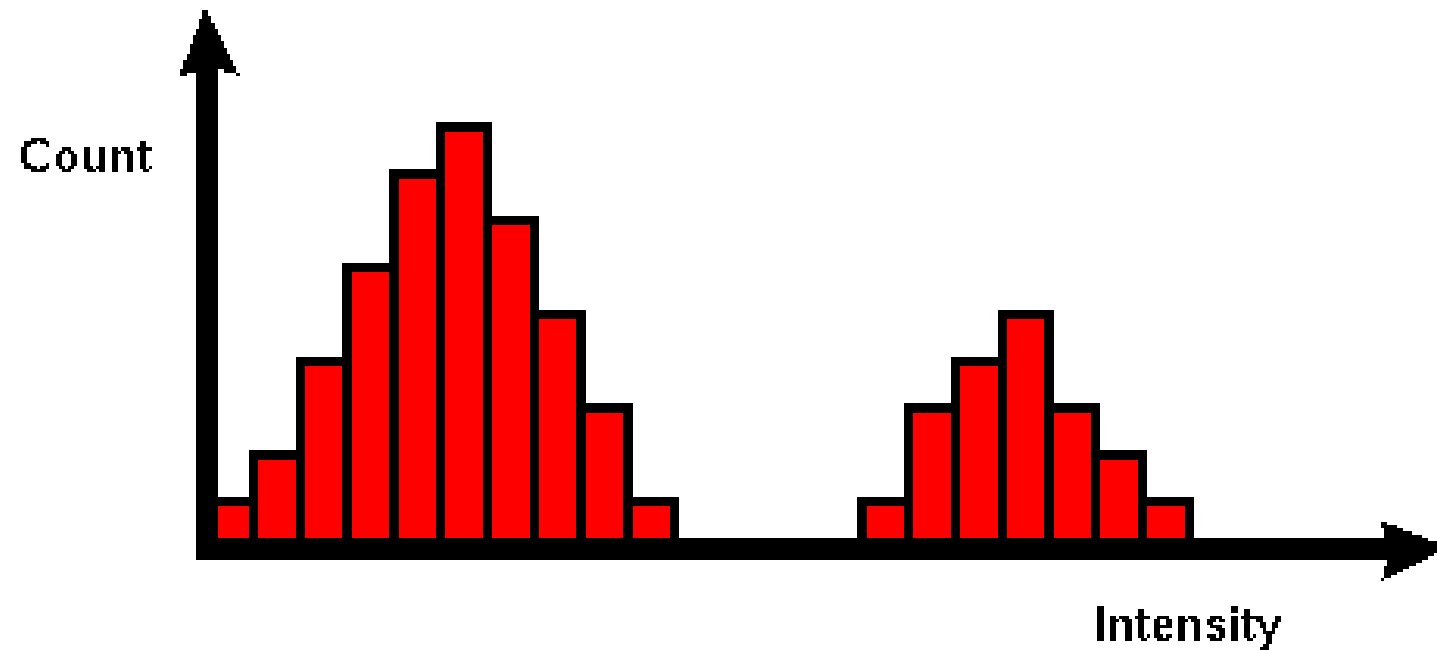
Bit depth



Bit depth	Binary numbers	No. of possible combinations
1-bit	0/1	2
2-bit	00/01/10/11	$2^2 = 4$
4-bit	0000/0001/0010/0011 ~ 1110/1111	$2^4 = 16$
8-bit	00000001/00000010 ~ 11111110/11111111	$2^8 = 256$
10-bit	0000000000 ~ 1111111111	$2^{10} = 1024$
12-bit	000000000000 ~ 111111111111	$2^{12} = 4096$
14-bit	00000000000000 ~ 11111111111111	$2^{14} = 16,384$
16-bit	0000000000000000 ~ 1111111111111111	$2^{16} = 65,536$

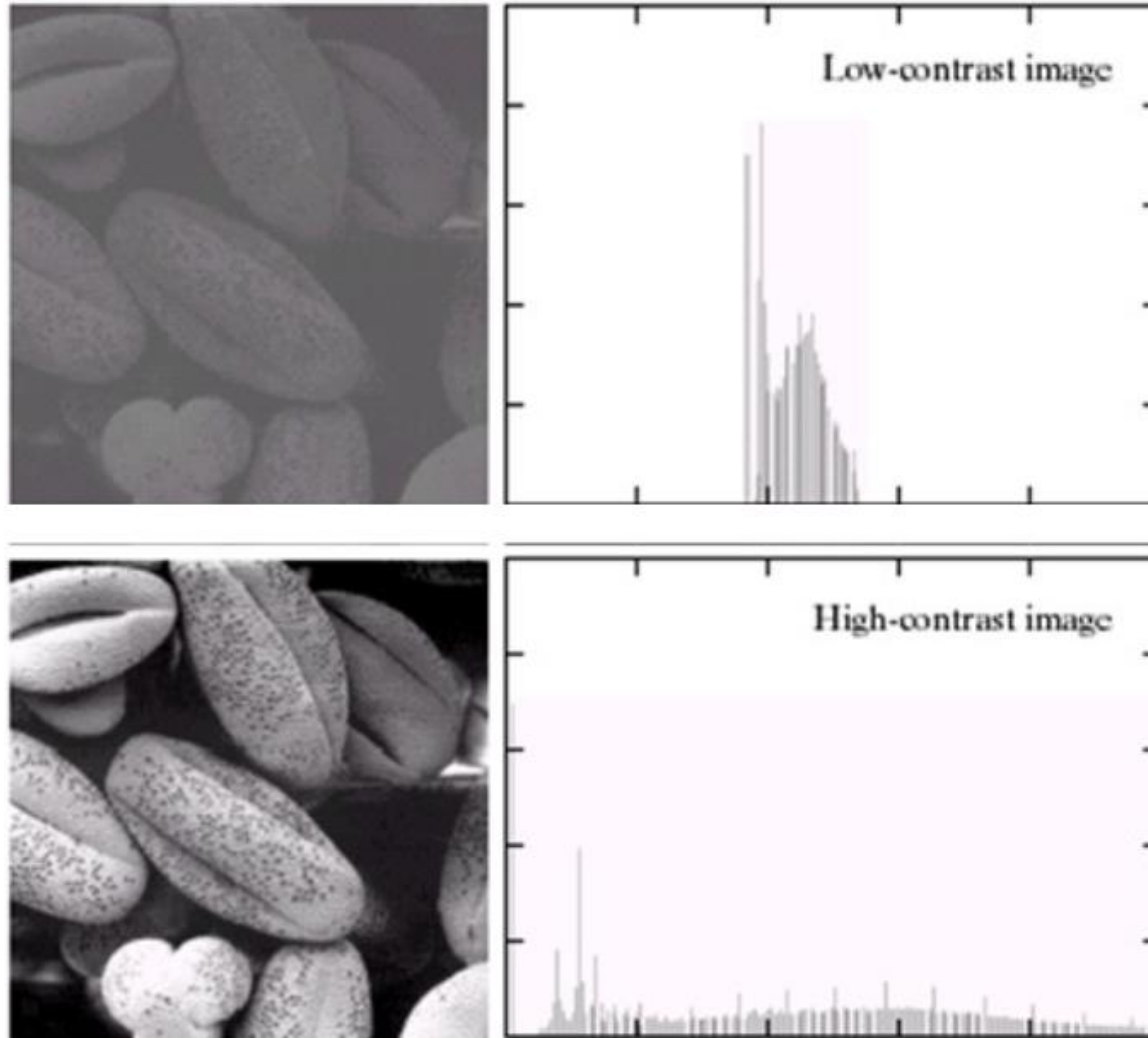


The pixel intensity histogram



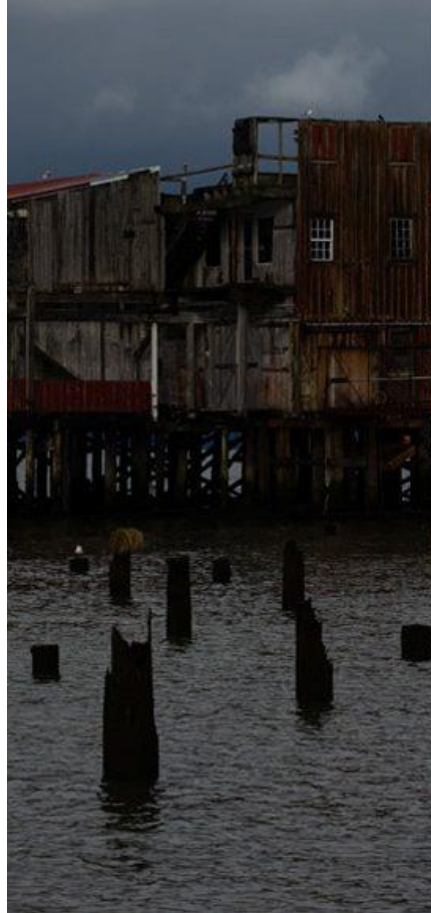
157	153	174	168	150	182	129	157	172	163	155	156
155	182	163	74	75	62	39	17	119	210	180	154
180	180	50	14	34	6	10	39	48	106	159	181
206	100	5	124	131	111	120	204	166	15	56	180
194	58	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	58	74	206
188	88	179	209	185	215	211	158	136	75	20	169
189	97	165	84	10	168	134	11	51	62	22	148
199	168	191	193	198	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	135	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	226	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

The pixel intensity histogram



Exposure times and overexposure

Low exposure time



High exposure time



Exposure times and overexposure

Low exposure time

High exposure time



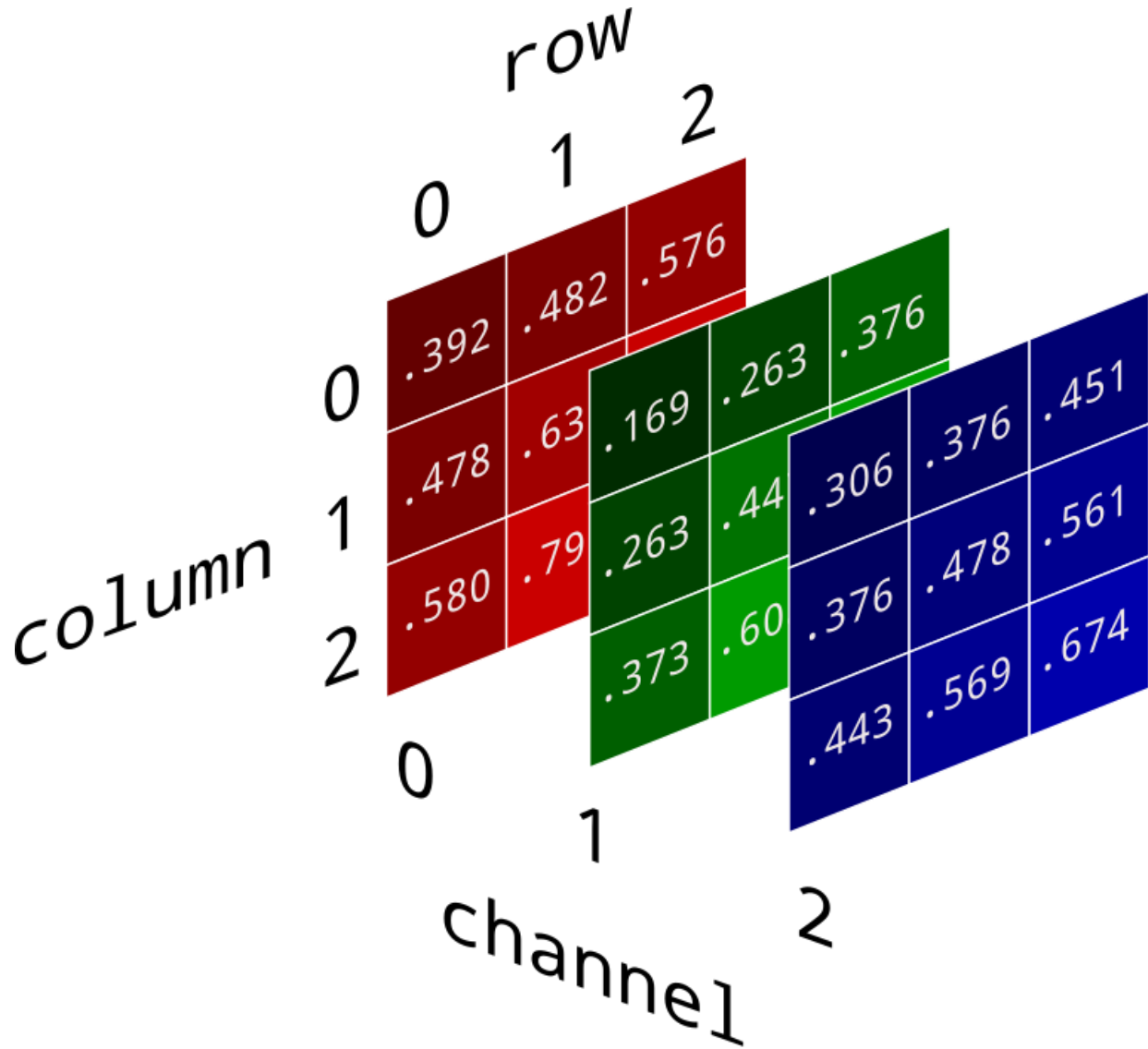
UNDEREXPOSED



OVEREXPOSED

**Image formats,
preserving the information**

Multiplexed images



Multiplexed images

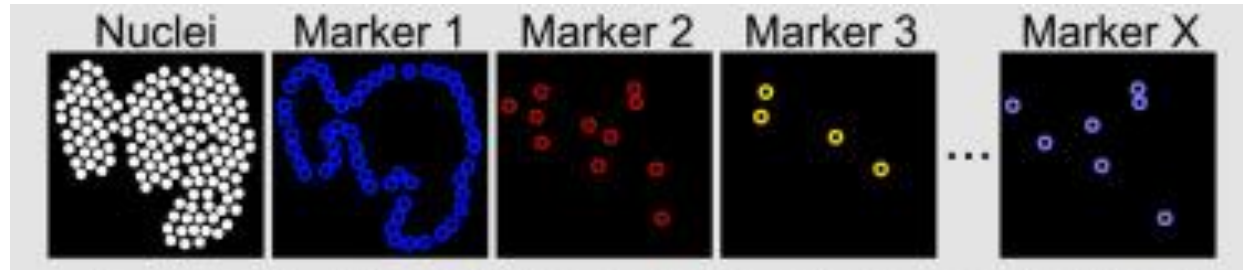
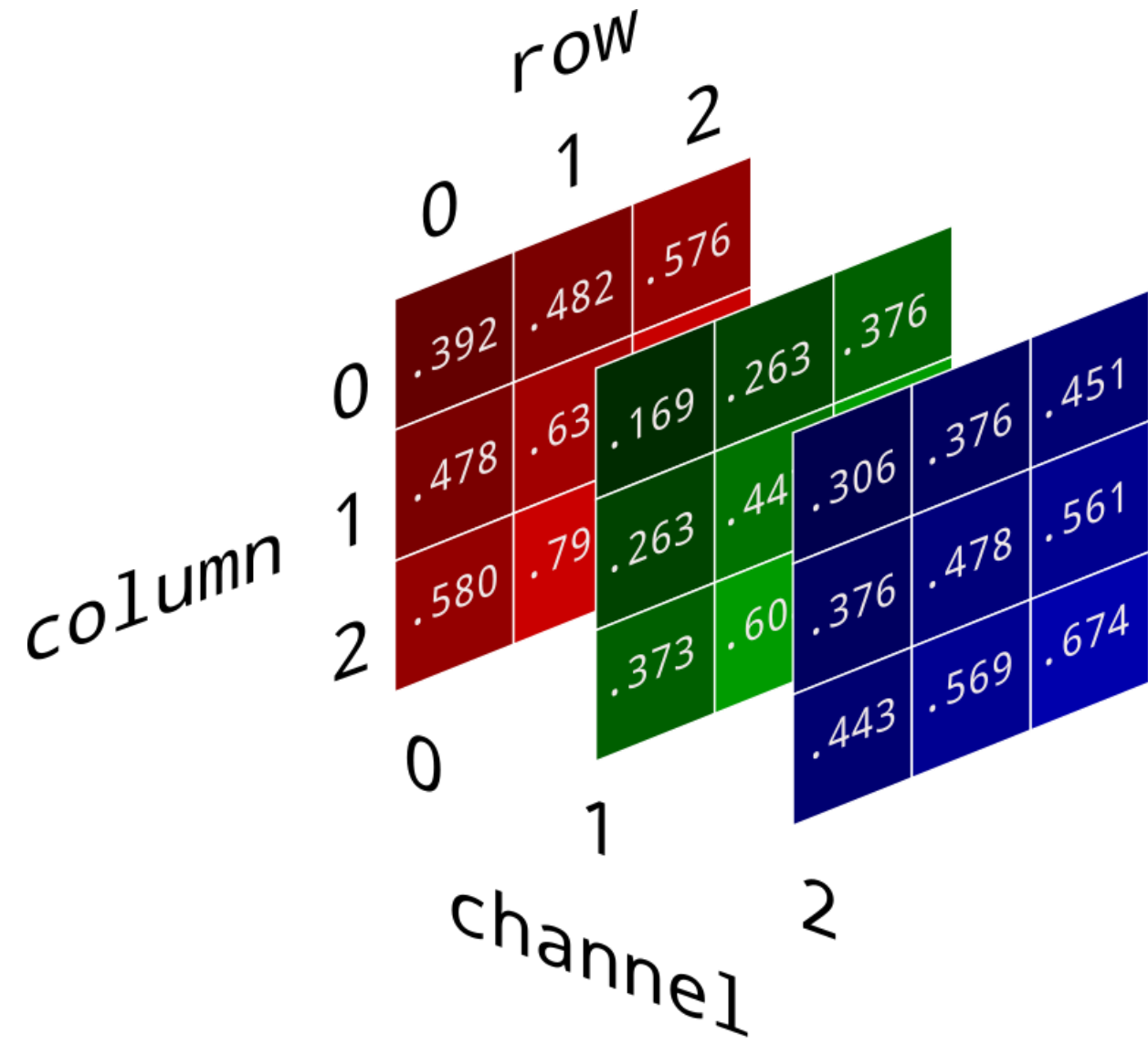


Image formats

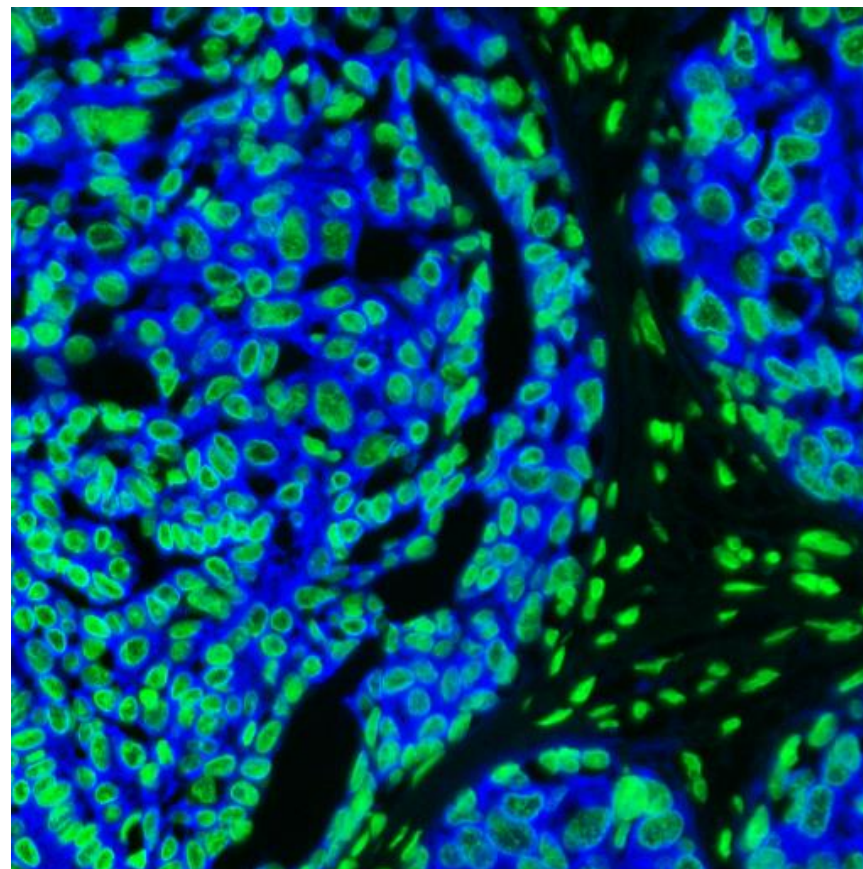
Format	Compression	Bit Depth	Metadata	Multi-Page	Size
TIFF	Lossless	8-bit, 16-bit, 32-bit	Extensive	Yes	Large
PNG	Lossless	8-bit, 16-bit	Moderate	No	Medium
JPEG	Lossy	8-bit	Limited	No	Small

OME-TIFF

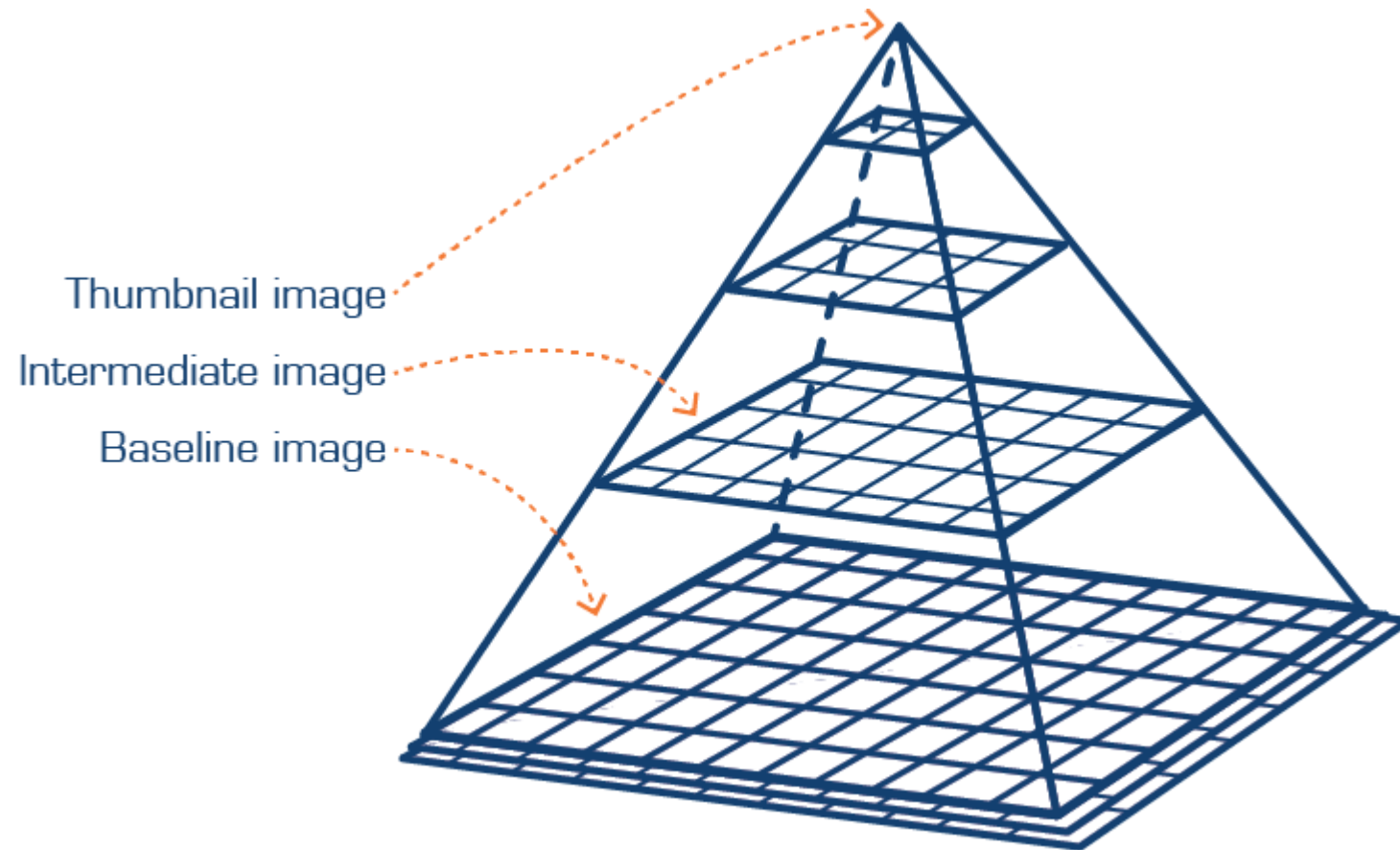
Open Microscopy Environment - Tagged Image File Format



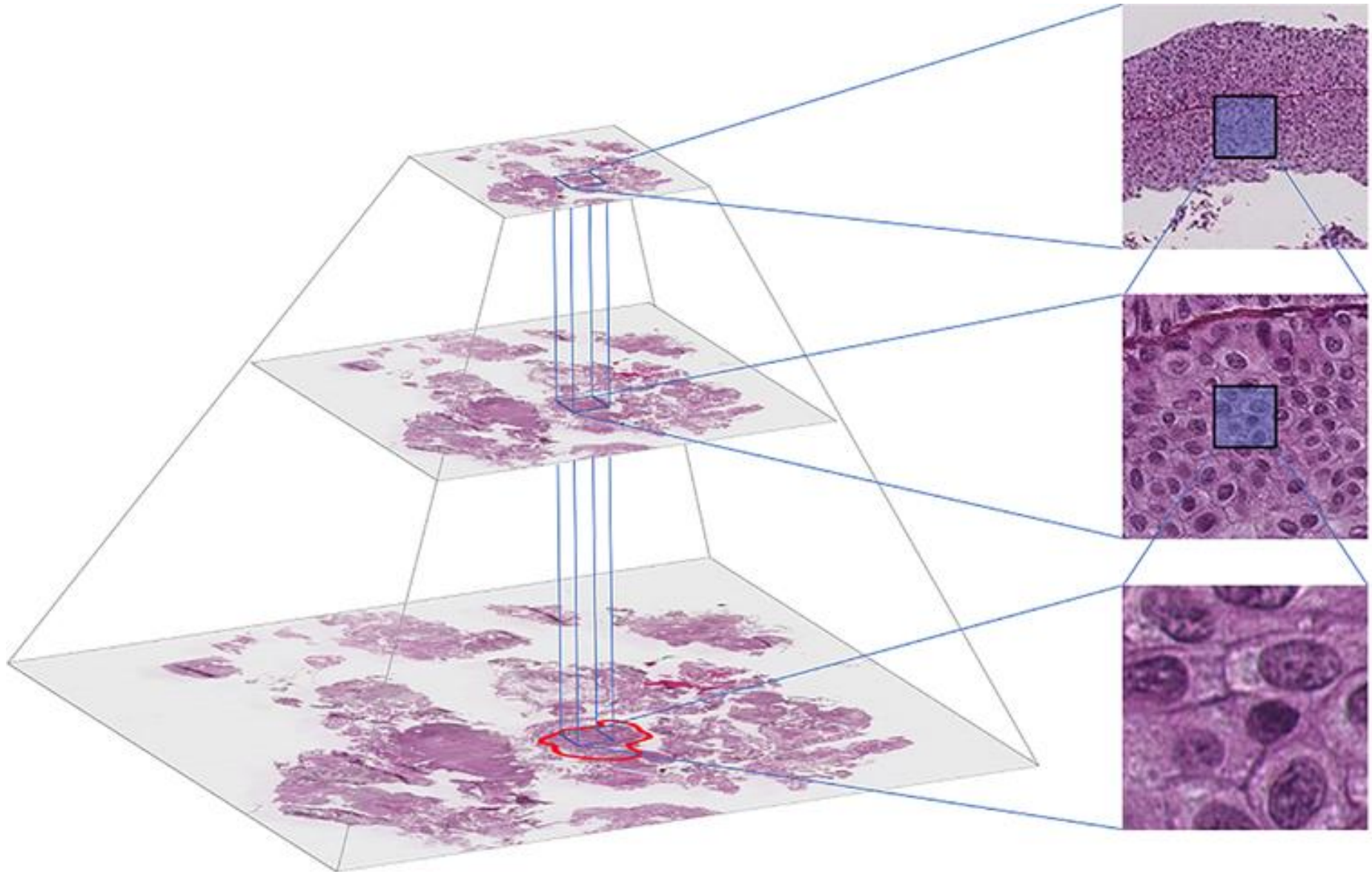
- ✓ Ensures **reproducibility**
- ✓ Facilitates **automated image analysis**
- ✓ Excellent **metadata preservance**
- ✓ Works across **different microscope brands & software**



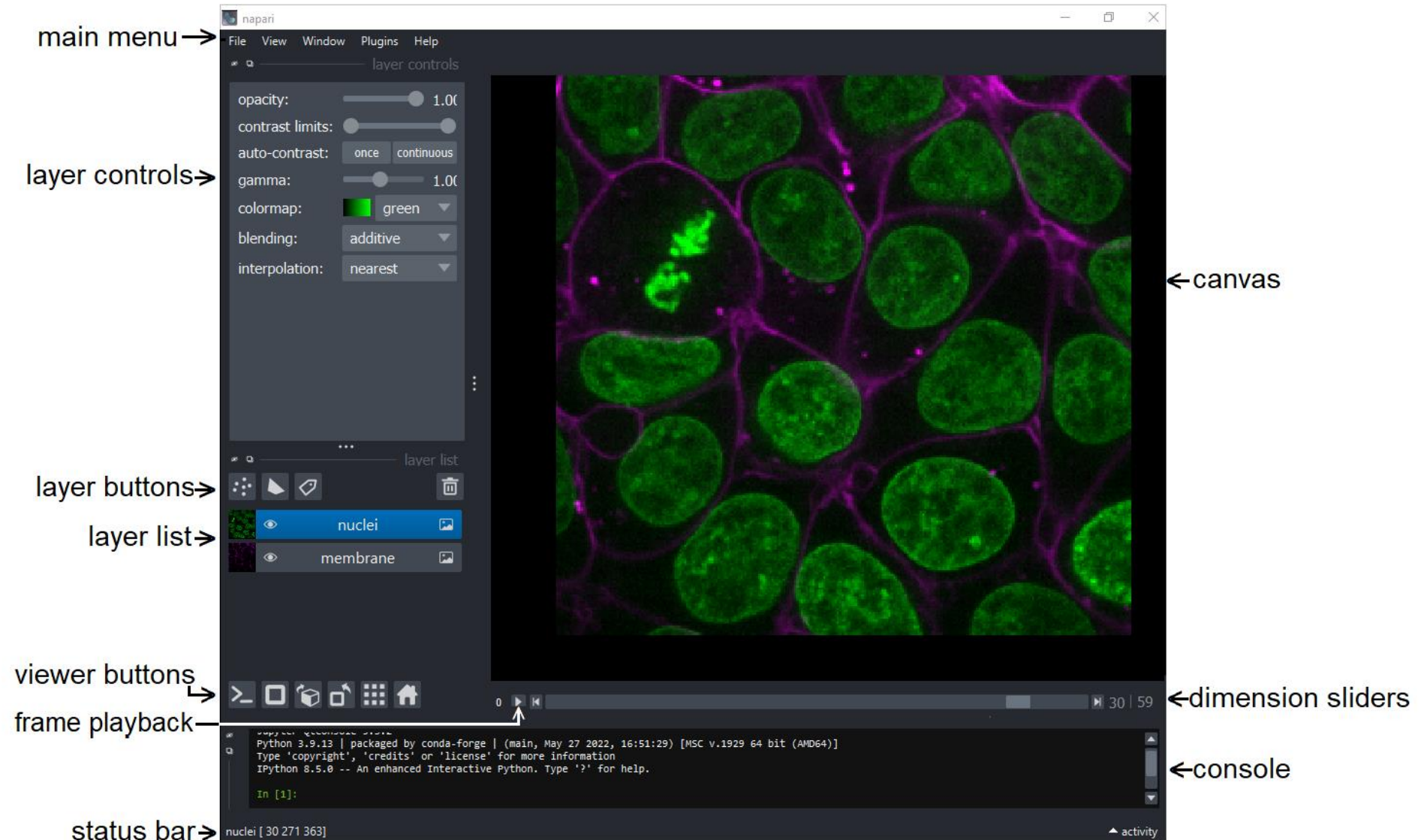
Pyramidal TIFF

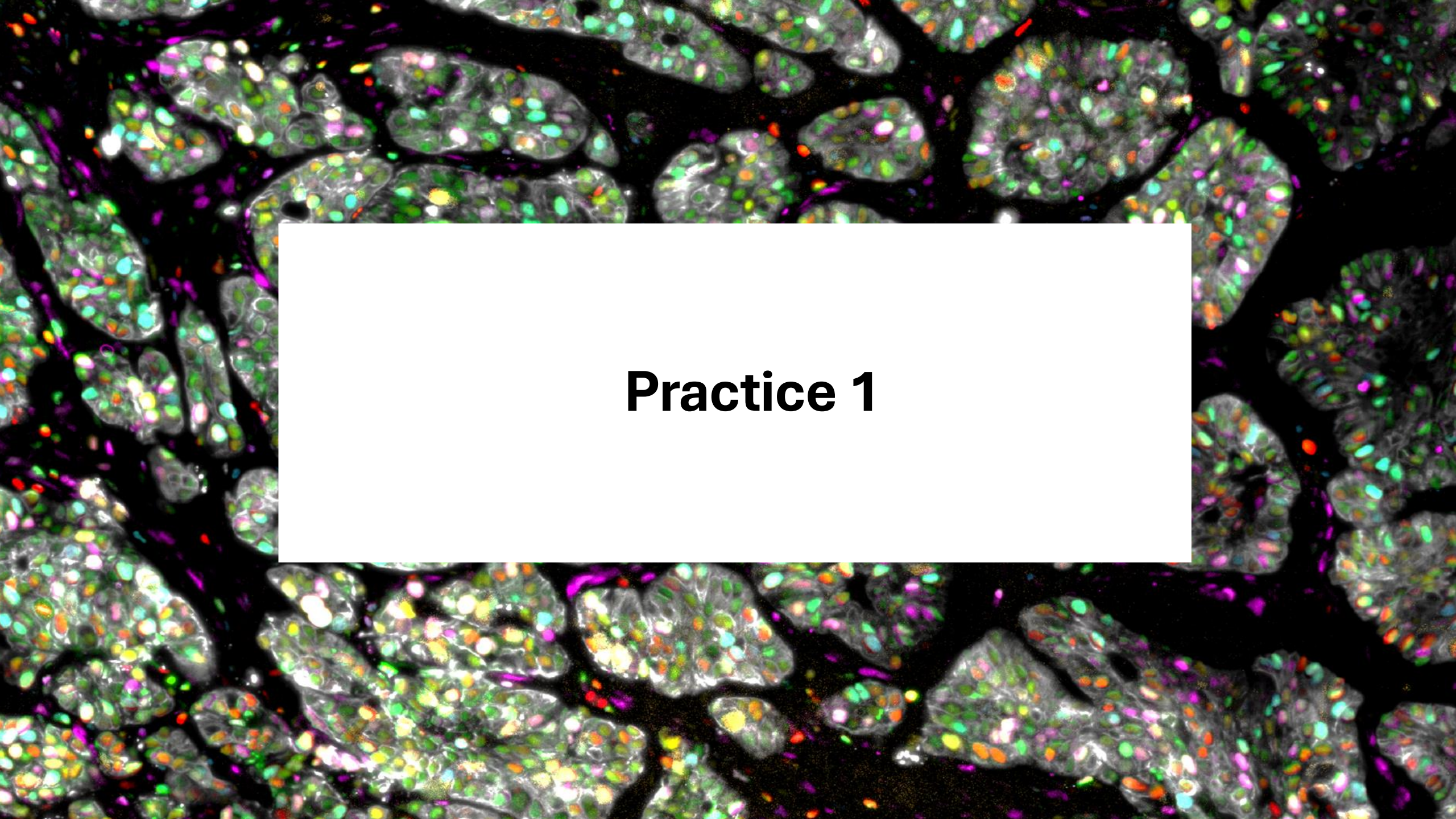


Pyramidal TIFF



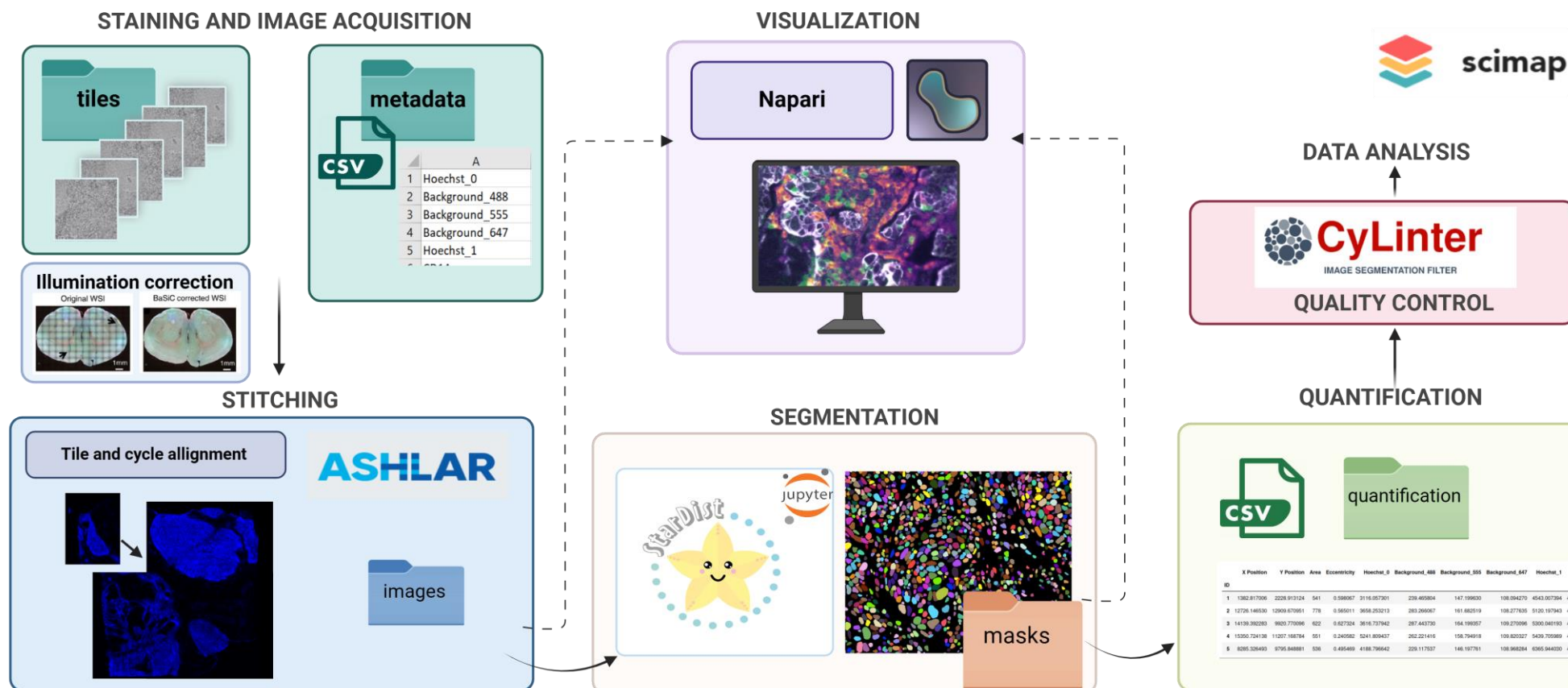
Napari





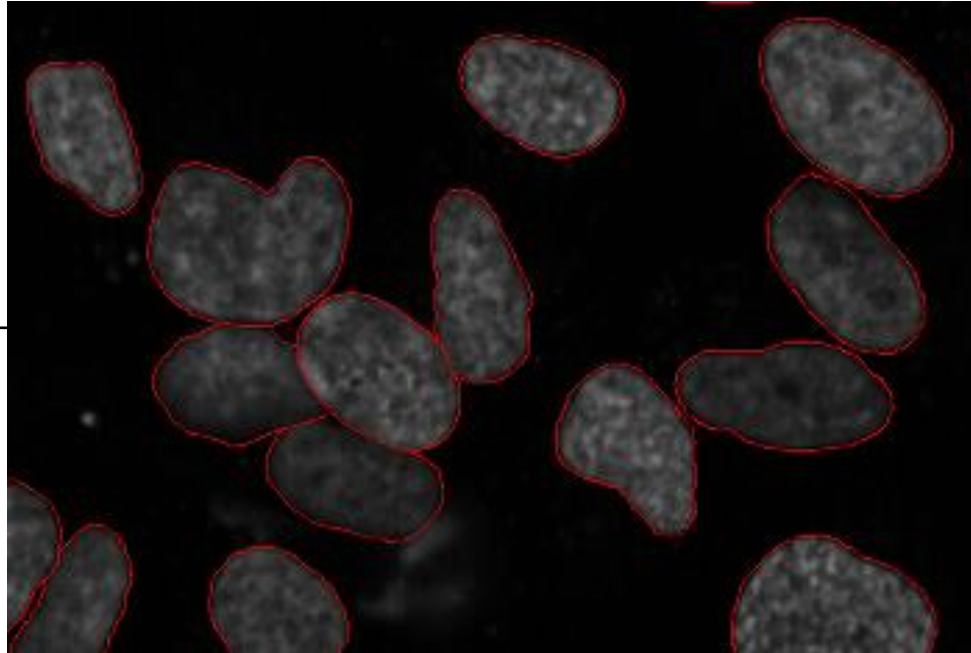
Practice 1

Image processing workflow



3. Cell segmentation

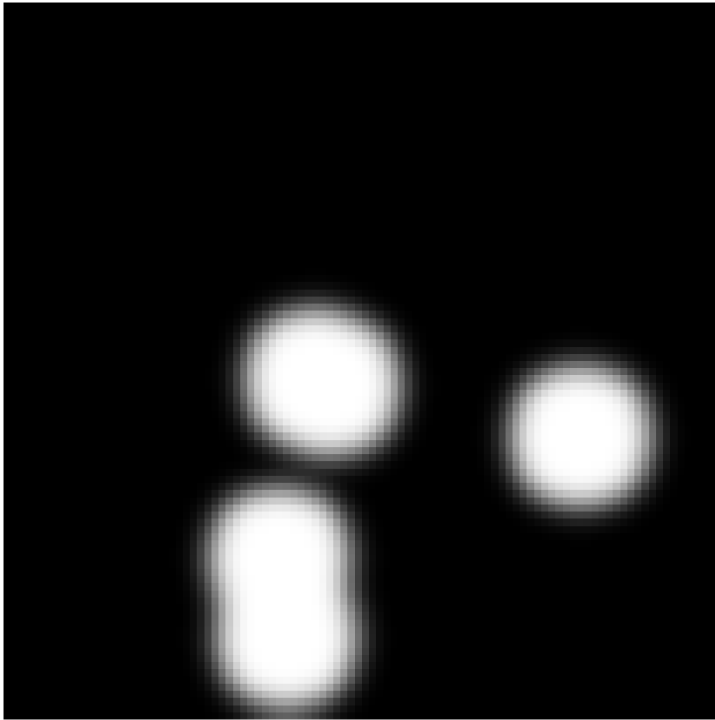
Why do we segment cells?



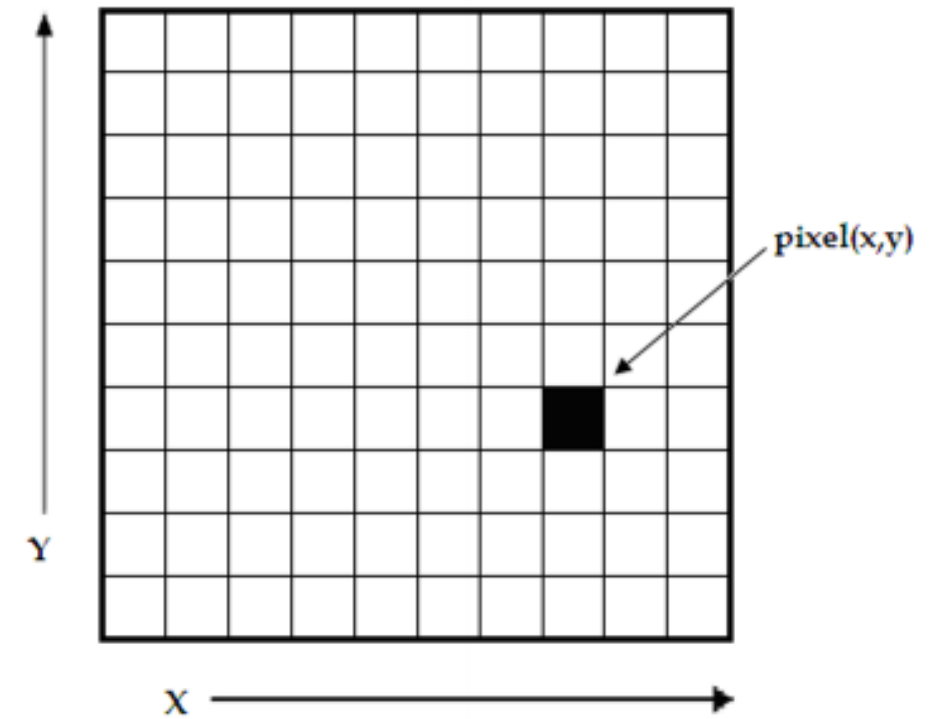
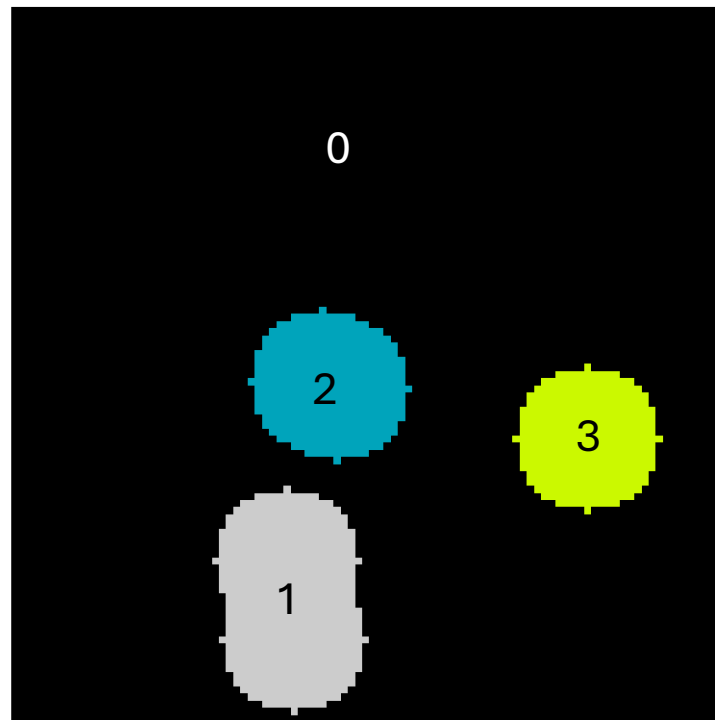
CellID	X position	Y position	Marker
1	48578.02	12079.58	291.29
2	12508.93	18347.84	840.30
3	41332.14	16543.74	299.86

Measuring pixel intensities in a given channel

DNA channel

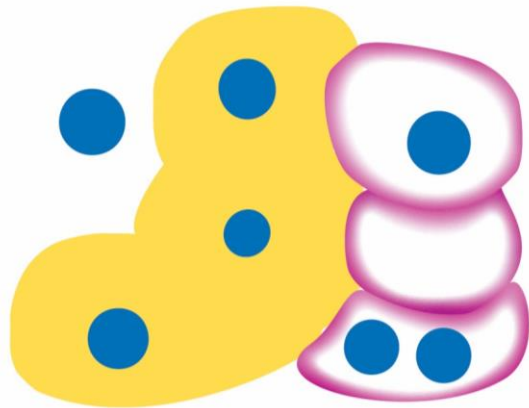


Mask



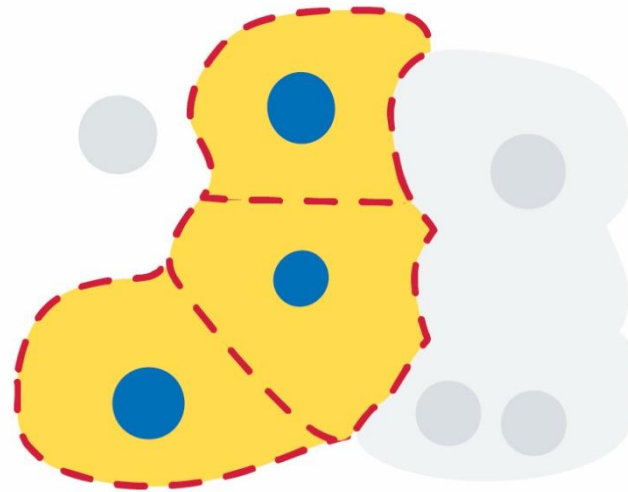
Strategies for cell segmentation

Marker
distribution



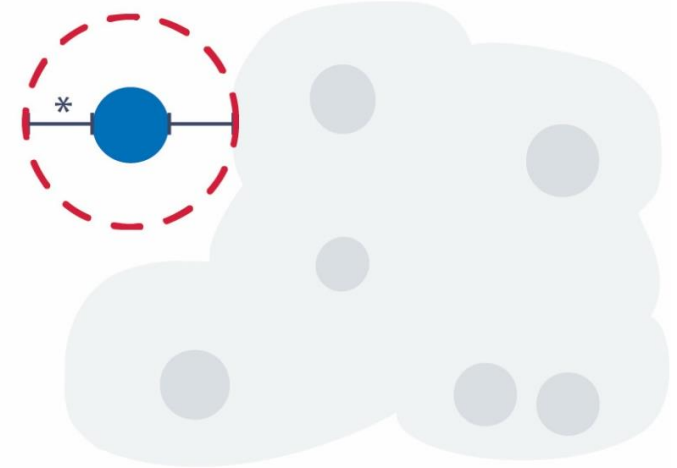
- Nuclear
- Interior
- Boundary

Nuclear
segmentation

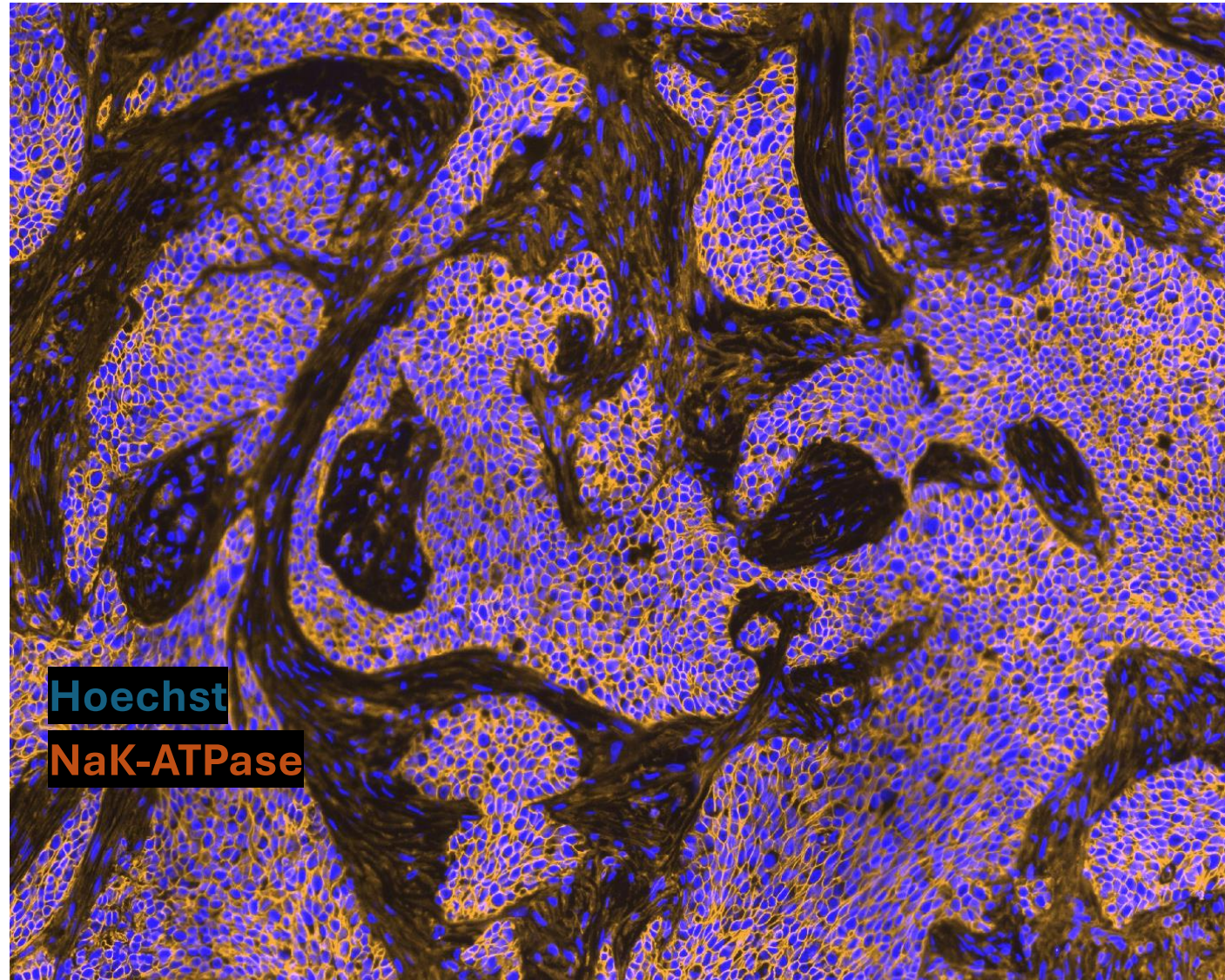


Membrane
segmentation

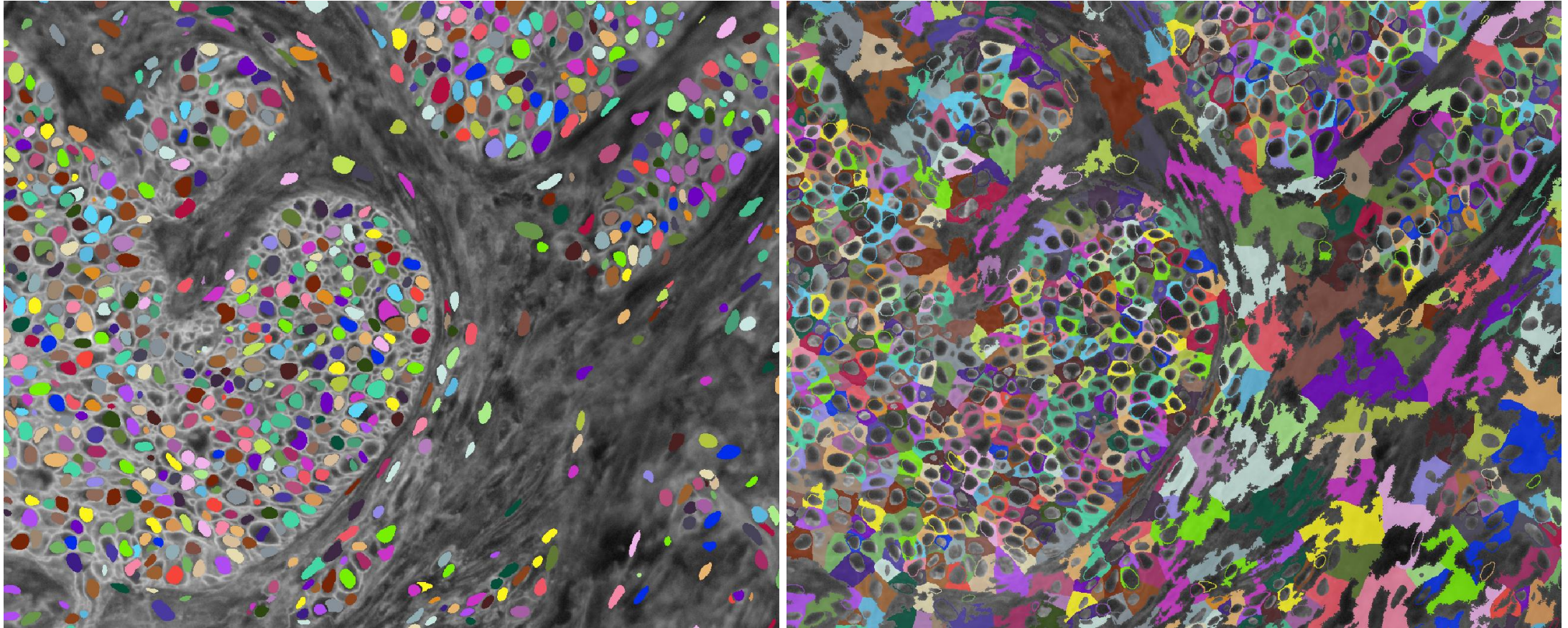
Nuclear mask
expansion



Attempting membrane segmentation: possible challenges

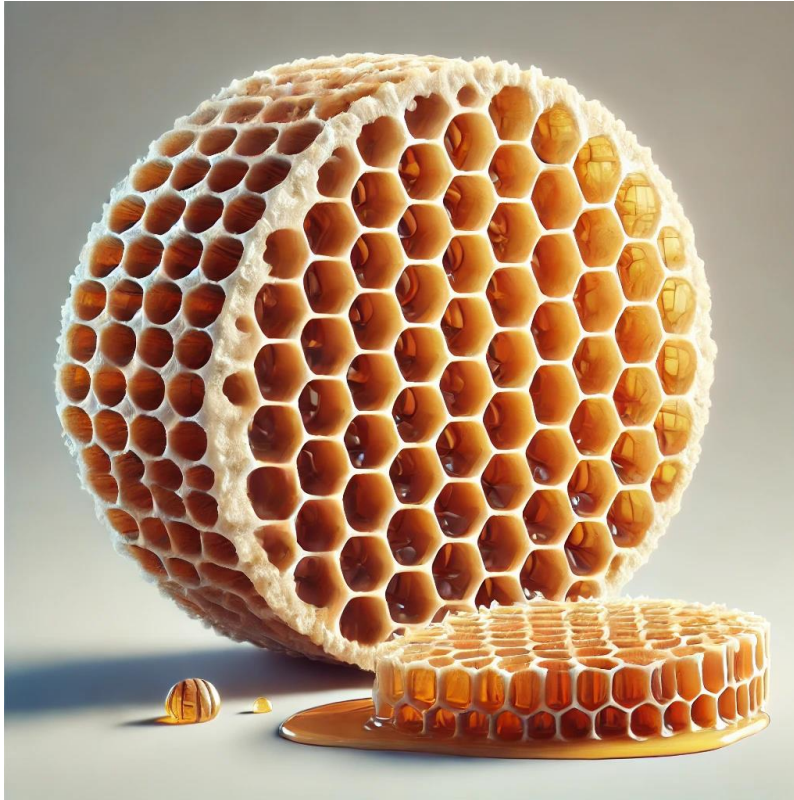


Attempting membrane segmentation: possible challenges

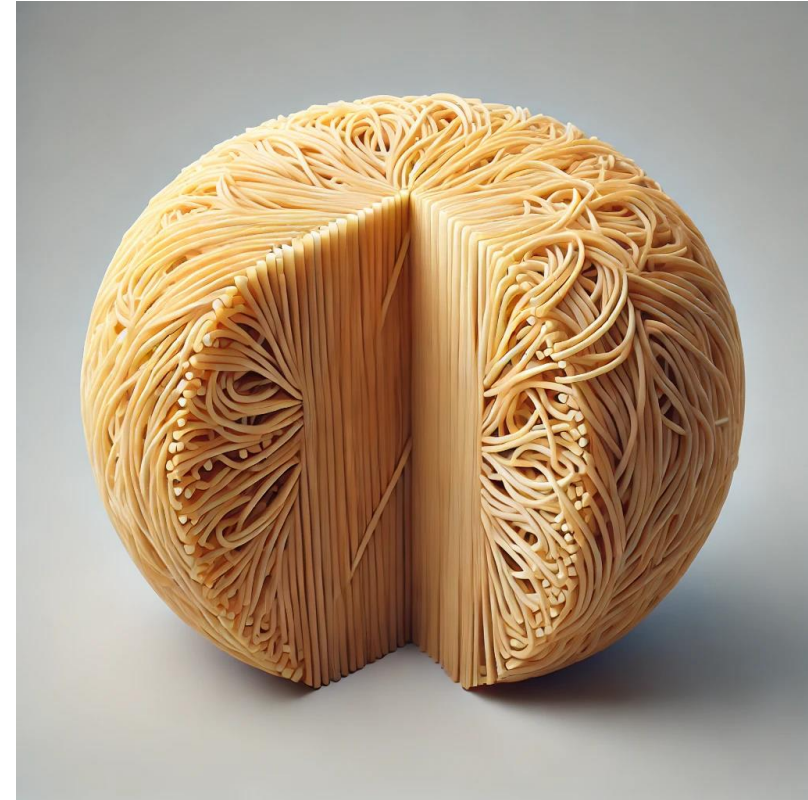


Attempting membrane segmentation: possible challenges

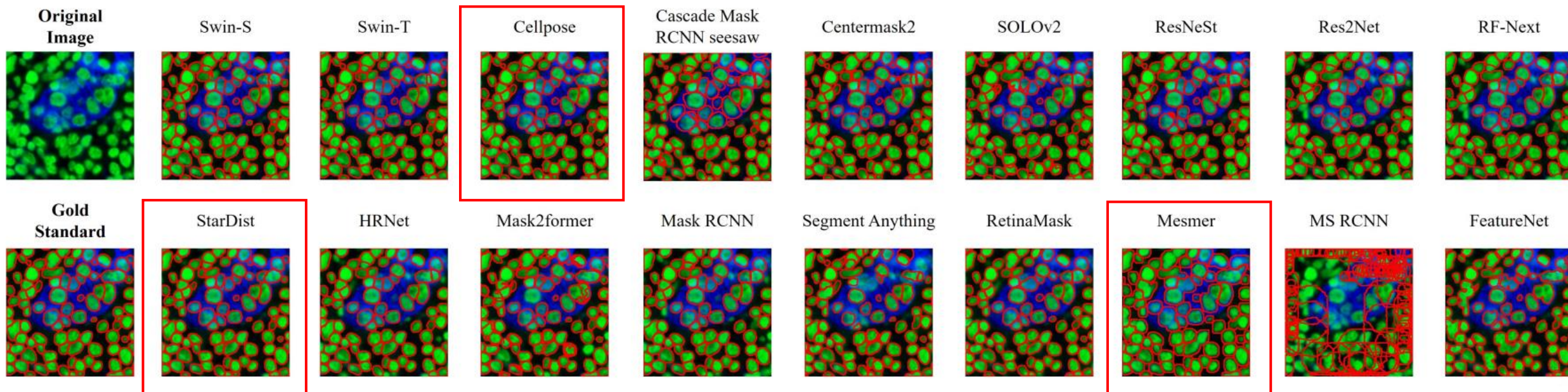
Epithelial cells



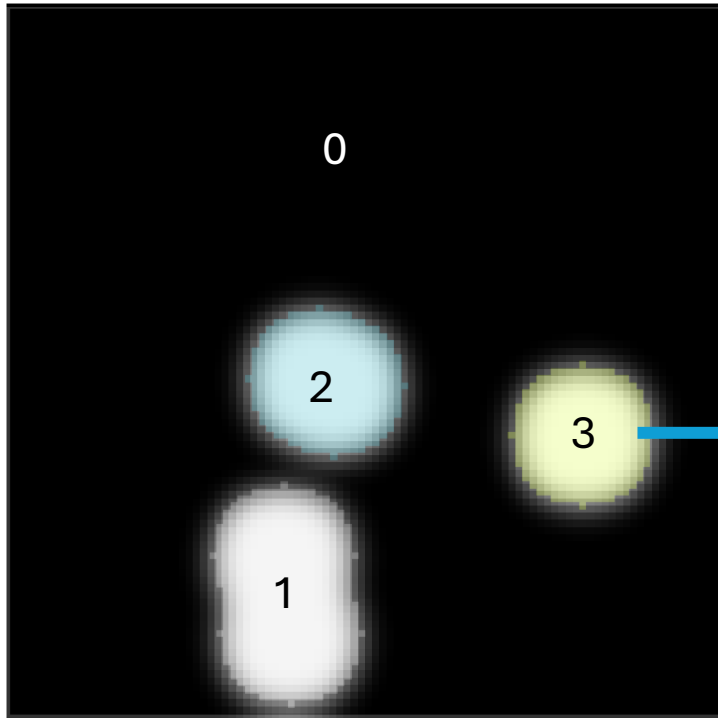
Stromal cells



We can choose among very different segmentation methods



The final quantified results



CellID	X position	Y position	Marker
1	48578.02	12079.58	291.29
2	12508.93	18347.84	840.30
3	41332.14	16543.74	299.86

Pixel value in the mask associated to the belonging of a specific cell

Mean or median value calculated from the **pixels under the mask** in a specific channel

What do intensities mean?

What do intensities mean?

$$I_{c,r,p} = \underbrace{B_{c,p}}_{\text{Biological}} \cdot \underbrace{T_{r,p}}_{\text{Technical}} \cdot \underbrace{S_{c,r,p}}_{\text{Spatial Context}} + \underbrace{\epsilon_{c,r,p}}_{\text{Computational Error}}$$

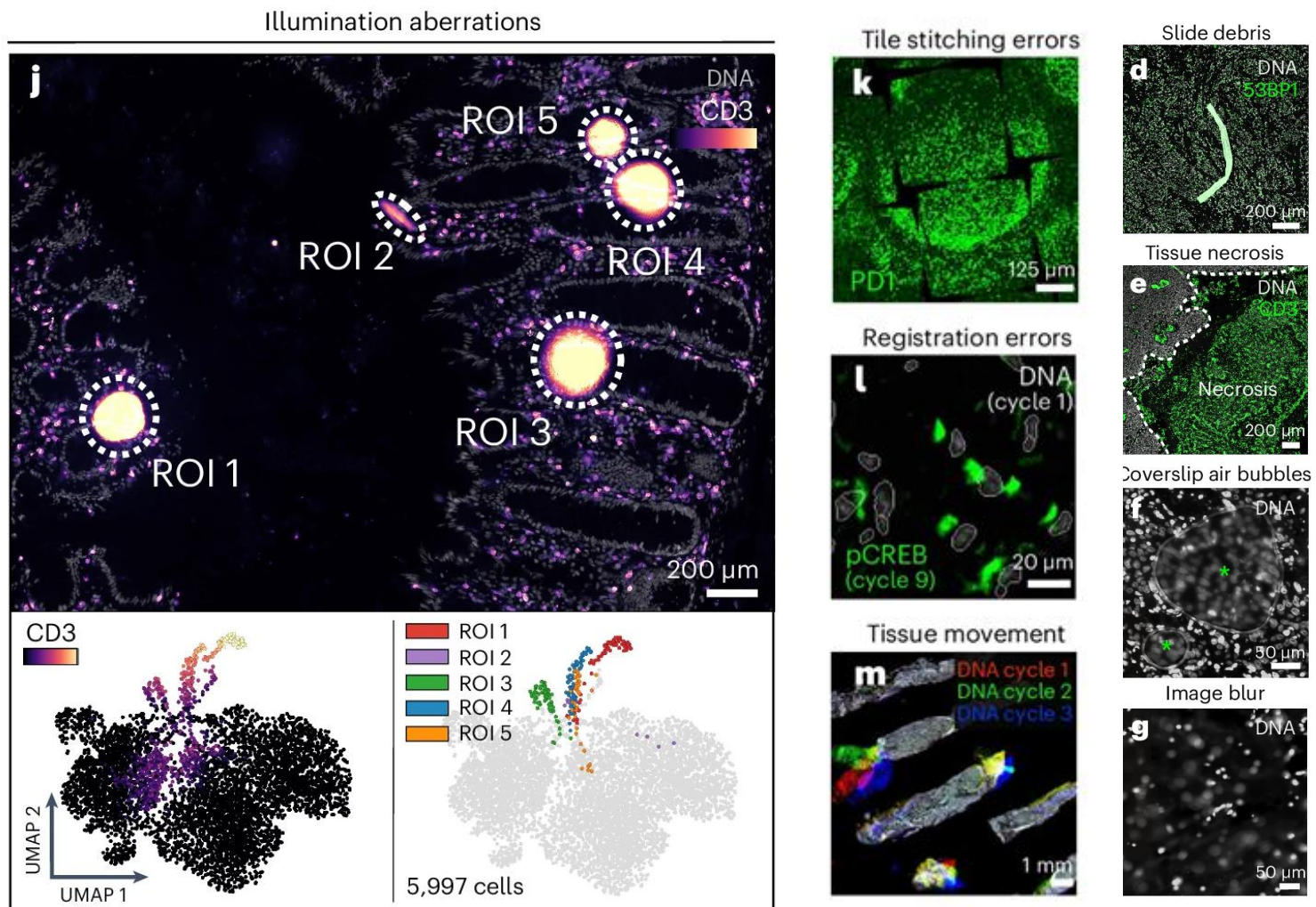
Quality control

**What changes
from scRNA-seq?**

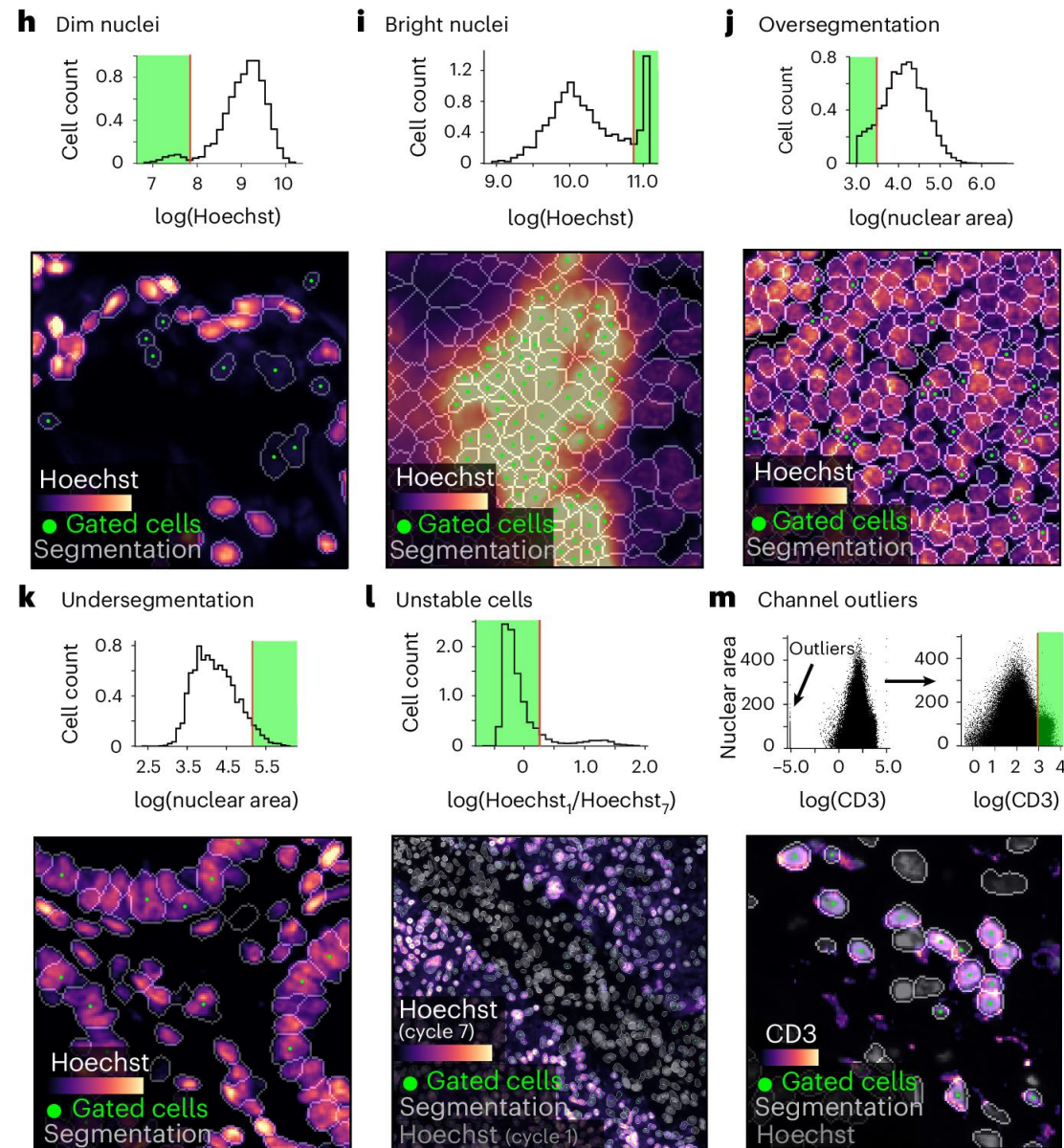
Which artifacts can we encounter?

Feature	scRNA-seq	t-CyCIF
Data Type	mRNA expression	Protein expression (immunofluorescence-based)
Spatial Information	Absent	Preserved
Number of Markers	1,000s–10,000s	~20–100
Signal Spillover	Minimal	Optical channel overlap, autofluorescence
Batch Effects	Reagent & platform variation	Staining, imaging batch variation
Cell Segmentation	Not required (cell barcodes used)	Image-based segmentation; error-prone
Quantification	Digital counts (UMIs)	Analog fluorescence intensity
QC Metrics Examples	Mitochondrial gene %; total UMI count; doublet rate	Signal-to-noise ratio; channel bleed-through; alignment error
Artifact Susceptibility	Cell stress, ambient RNA contamination	Autofluorescence, photobleaching, antibody non-specificity
Normalization Needs	Strong (due to dropouts, library size variation)	Moderate (for fluorescence intensity across samples)

Which artifacts can we encounter?



Which imprecisions do we introduce in the dataset?



Quality control recommendation



CyLinter

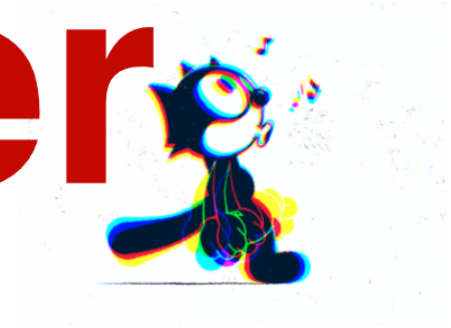
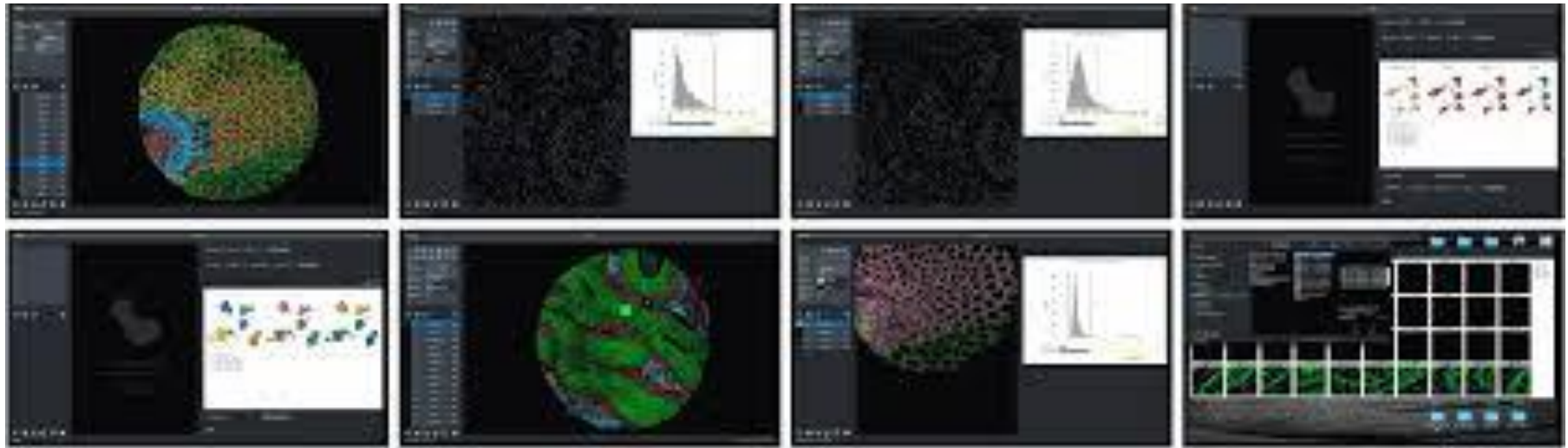
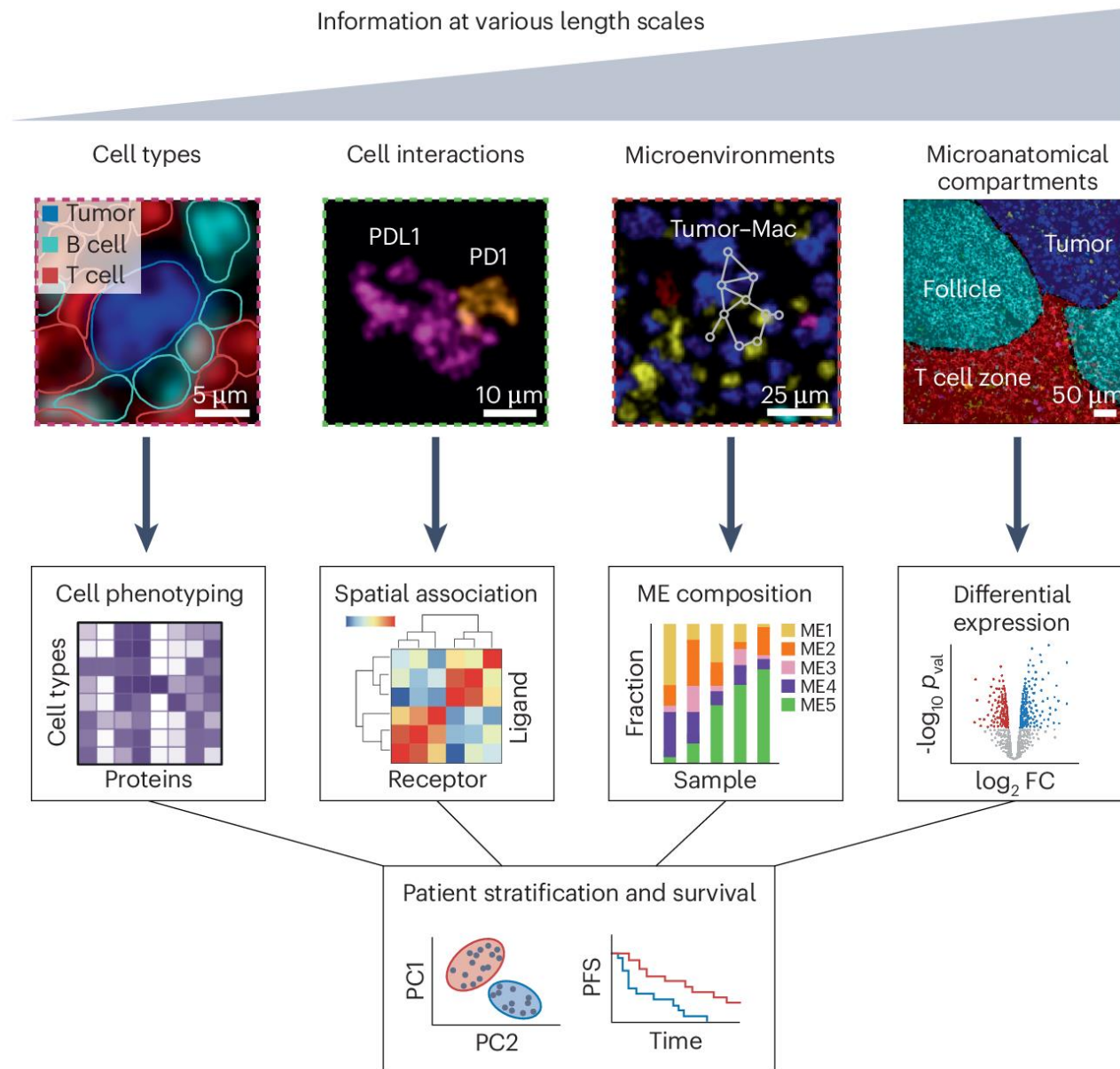
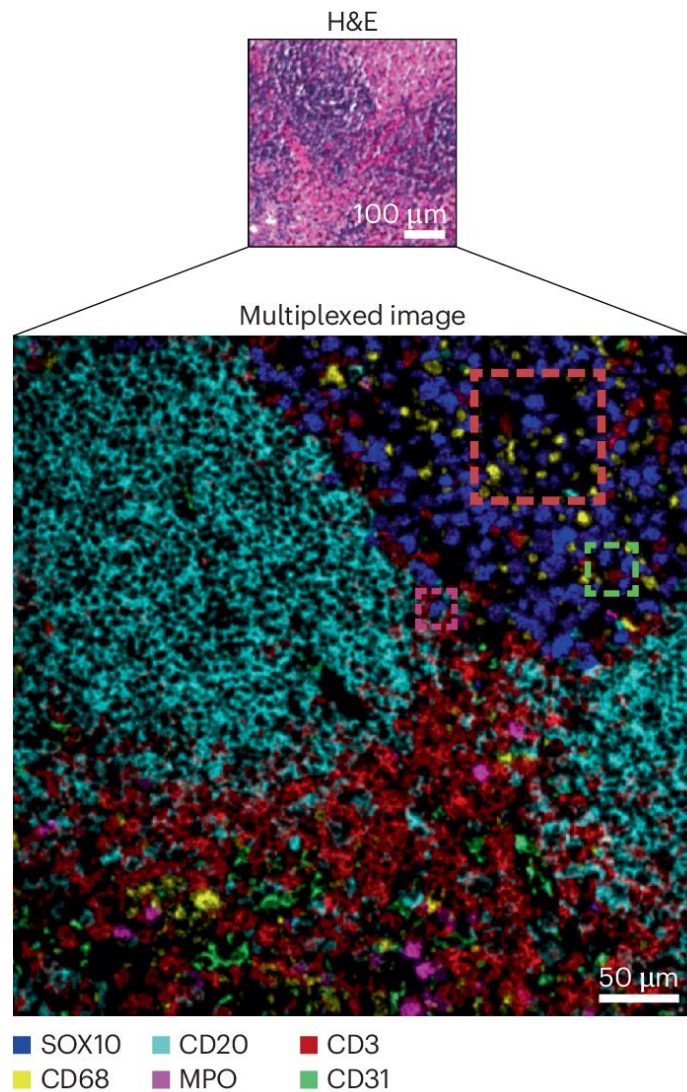
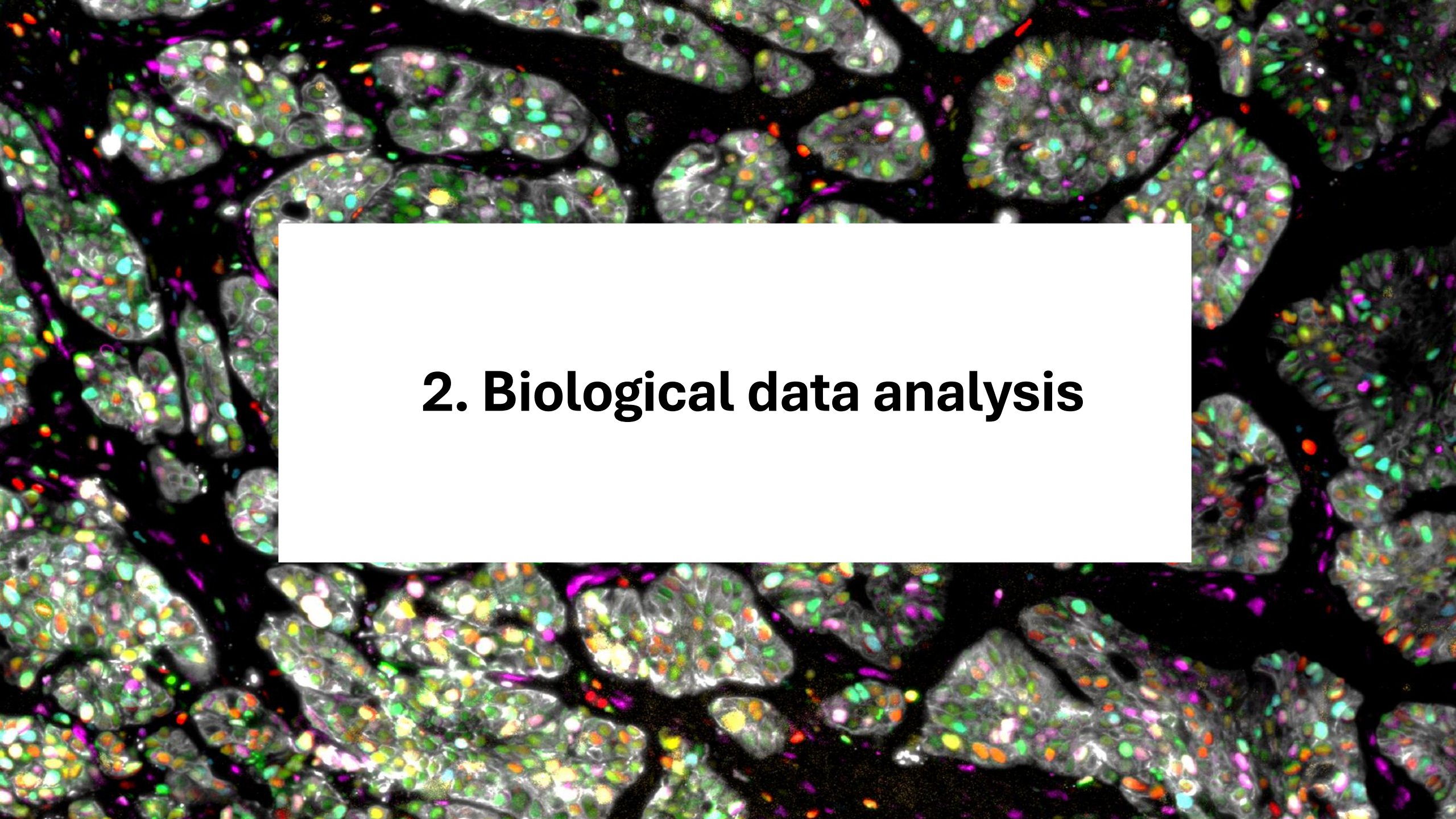


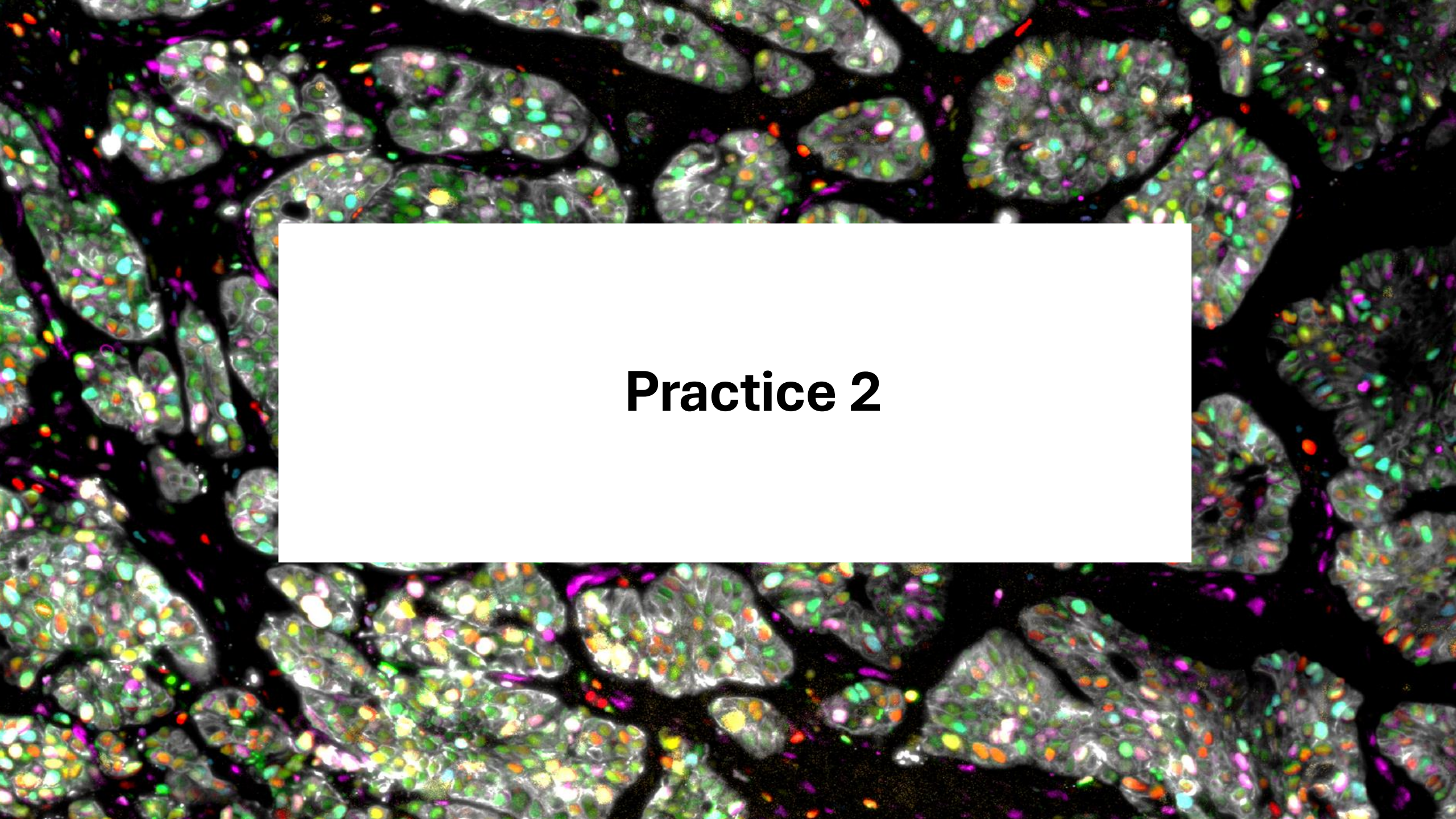
IMAGE SEGMENTATION FILTER





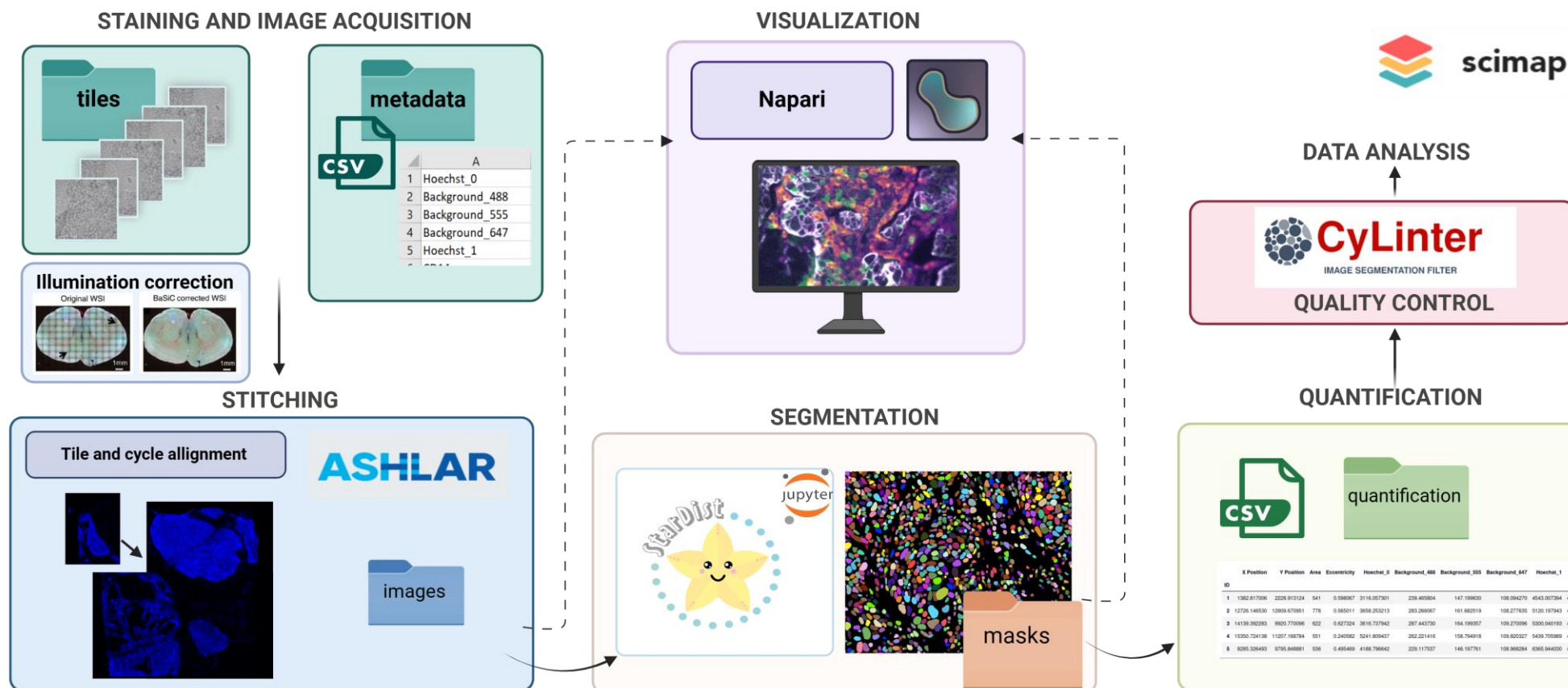
A microscopic image showing numerous cells, likely from a tissue sample, stained with various fluorescent dyes. The cells are irregular in shape and contain many small, brightly colored spots (green, red, blue, yellow, and purple) which represent different fluorescent markers or proteins. The background is dark, making the fluorescent cells stand out.

2. Biological data analysis



Practice 2

Image processing workflow



SCIMAP

Spatial Data
(.csv .h5ad)



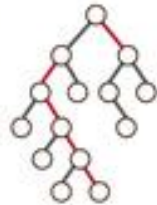
2D/3D

Cell type assignment

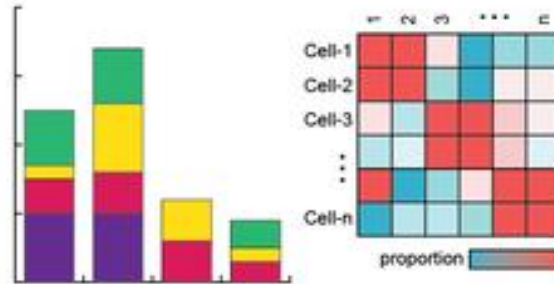


Clustering
based

Prior knowledge
based



Cellular frequency,
marker correlation, etc.



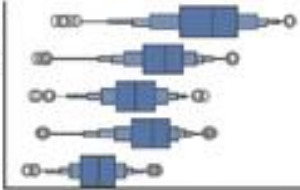
Spatial analysis

Build
neighbourhood
graph



Spatial
distances

cell type-1
cell type-2
cell type-n

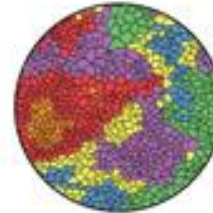


distance to cell type-x

Cell-cell
co-occurrence



Spatial LDA and
other neighbourhood
detection methods



Visualization

Voronoi diagram's
& other plotting tools

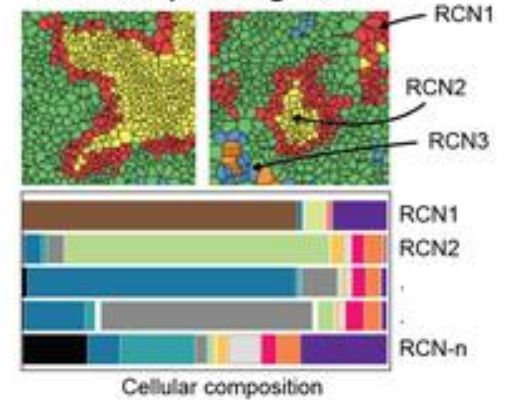
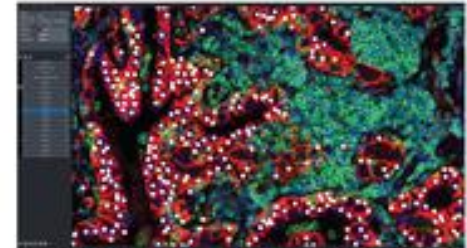
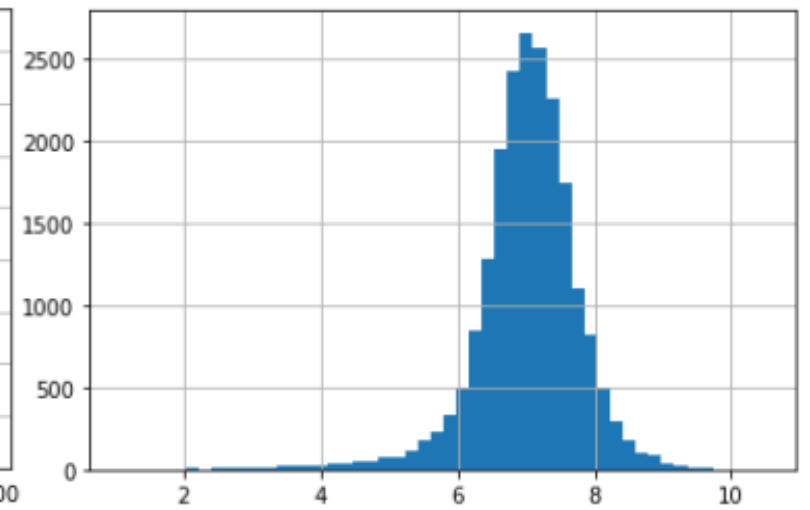
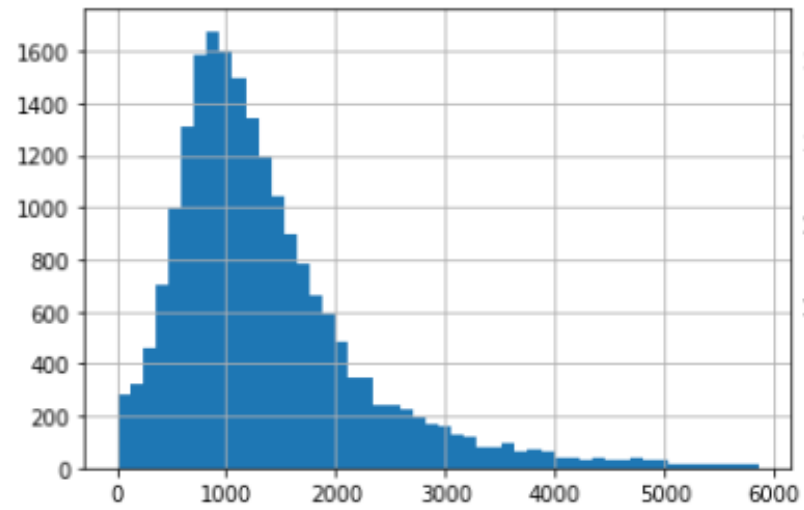
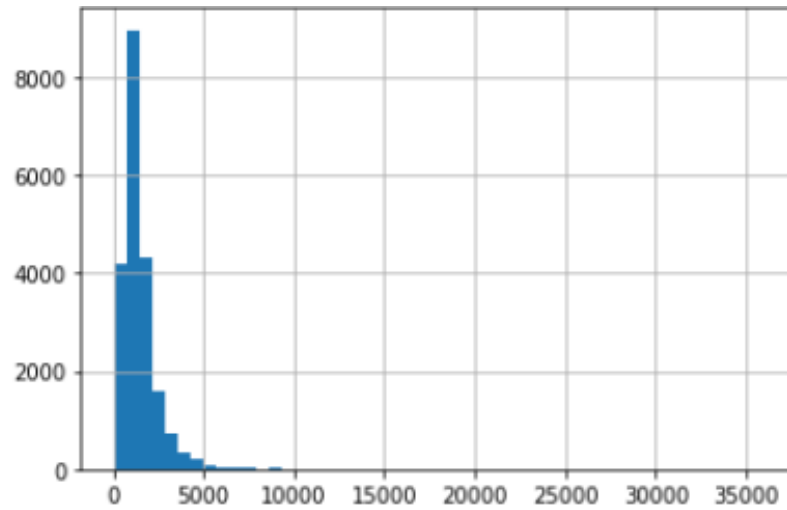


Image visualization: Napari



zarr, dask supported

Data normalization



How can we assign phenotypes to our cells?

CellID	X position	Y position	Marker	Phenotype
1	48578.02	12079.58	291.29	Cancer
2	12508.93	18347.84	840.30	CD8
3	41332.14	16543.74	299.86	Stromal
4	34043.81	24965.79	444.17	DC
5	21639.99	21359.70	514.44	CD4

**Phenotypes are a
social construct**

How can we assign phenotypes to our cells?

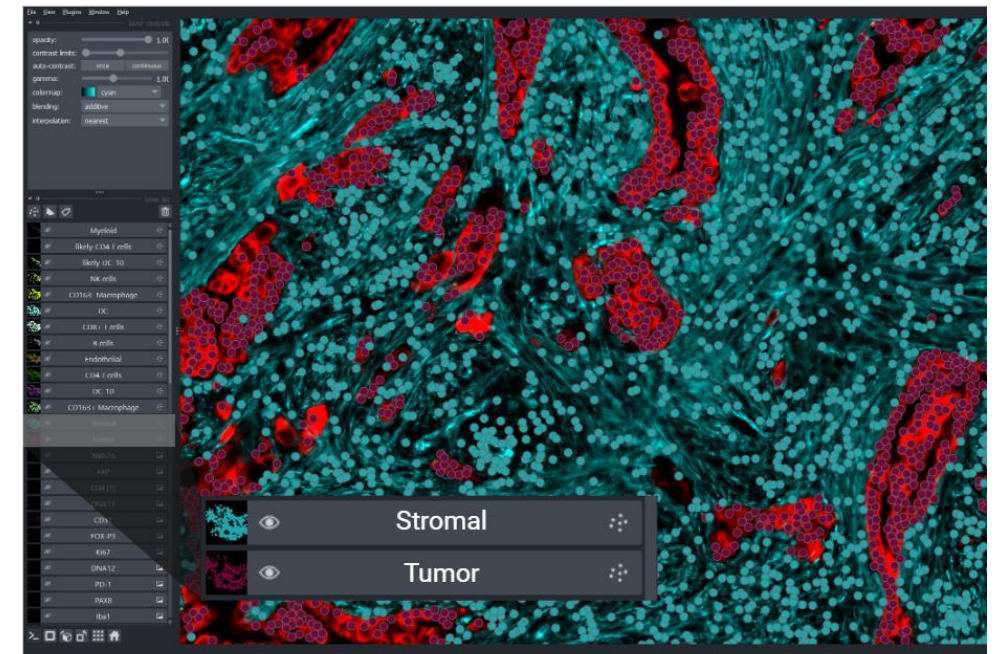
Phenotype markers

Cancer
PanCK
PAX8

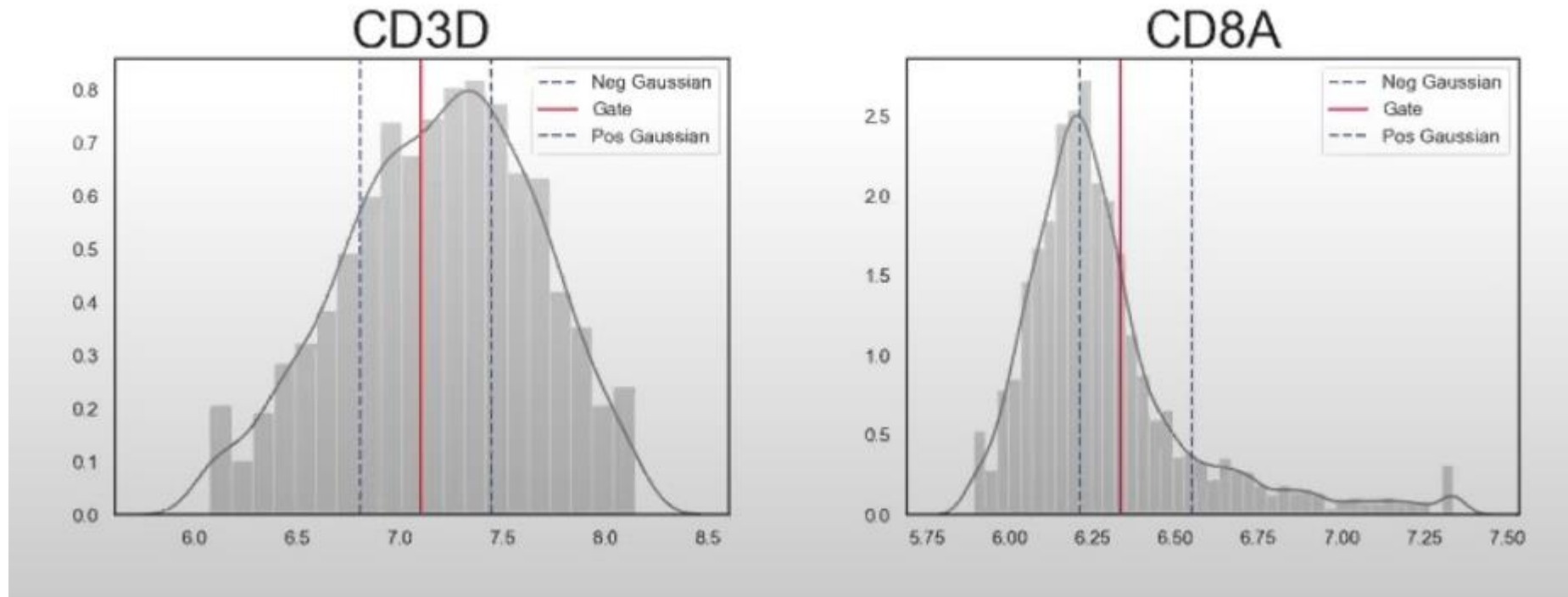
Immune
CD45
CD4
CD8a
CD20
FOXP3
NKG2a
Iba1
CD163
CD11c

Stroma
Vimentin
αSMA
FAP
CD31

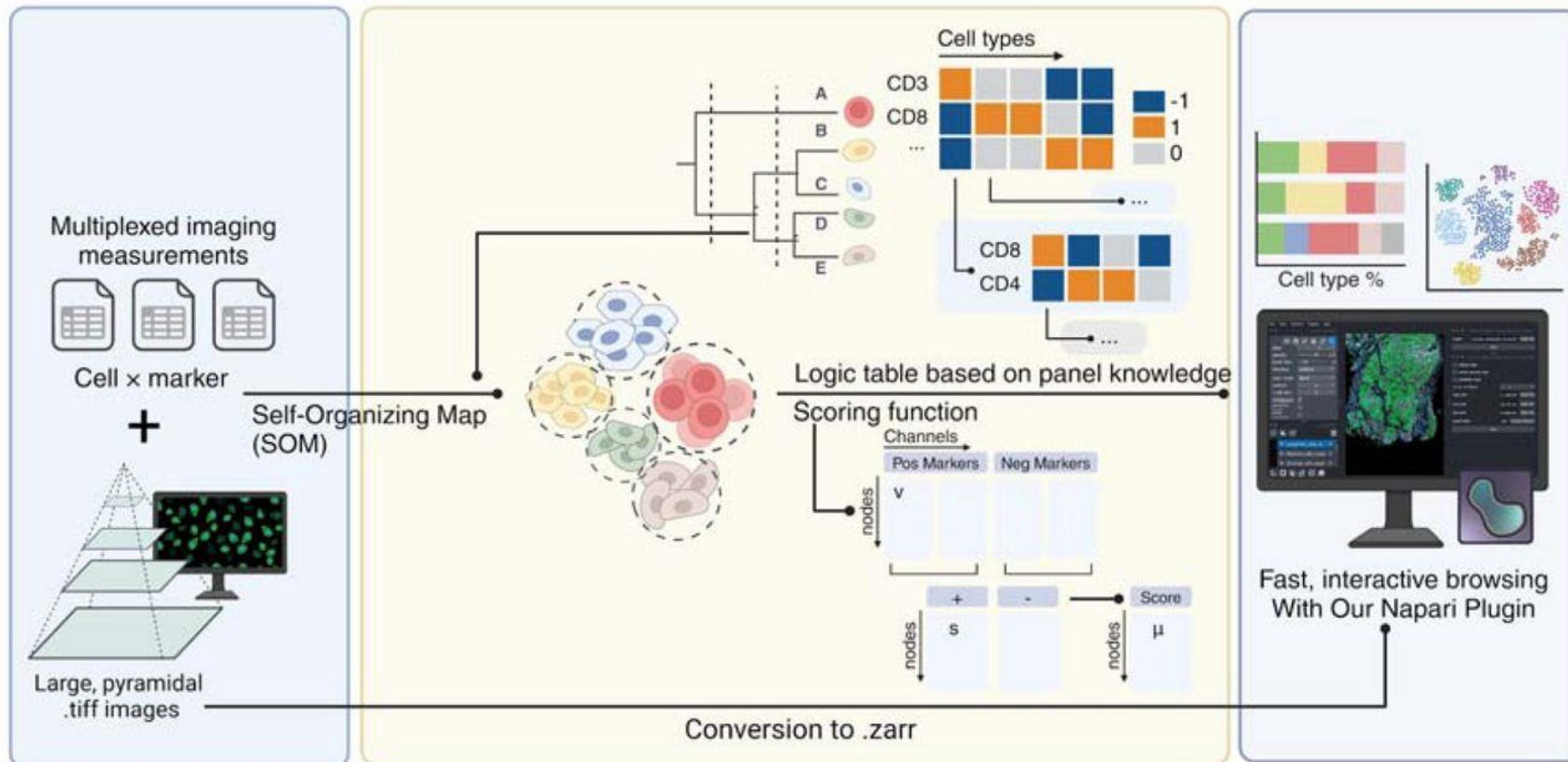
		PAX8	PanCK	Vimentin	αSMA	Iba1	CD11c	CD163	CD45	CD8a	CD4	CD20	CD31
all	Tumor	anypos	anypos										
all	Other			anypos	anypos	anypos			anypos	anypos			anypos
Other	Myeloid					anypos	anypos	anypos					
Other	Lymphoid								anypos	anypos			
Other	Stromal			anypos	anypos								anypos
Myeloid	CD163+ DC						pos	pos					
Myeloid	DC						pos	neg					
Myeloid	CD163- Macrophages					pos	allneg	allneg					
Myeloid	CD163+ Macrophages					allpos	neg	allpos					
Lymphoid	CD8+ T cells									pos			
all	CD4+ T cells								pos	neg	pos		
Lymphoid	B cells									neg		pos	
Stromal	Endothelial												pos



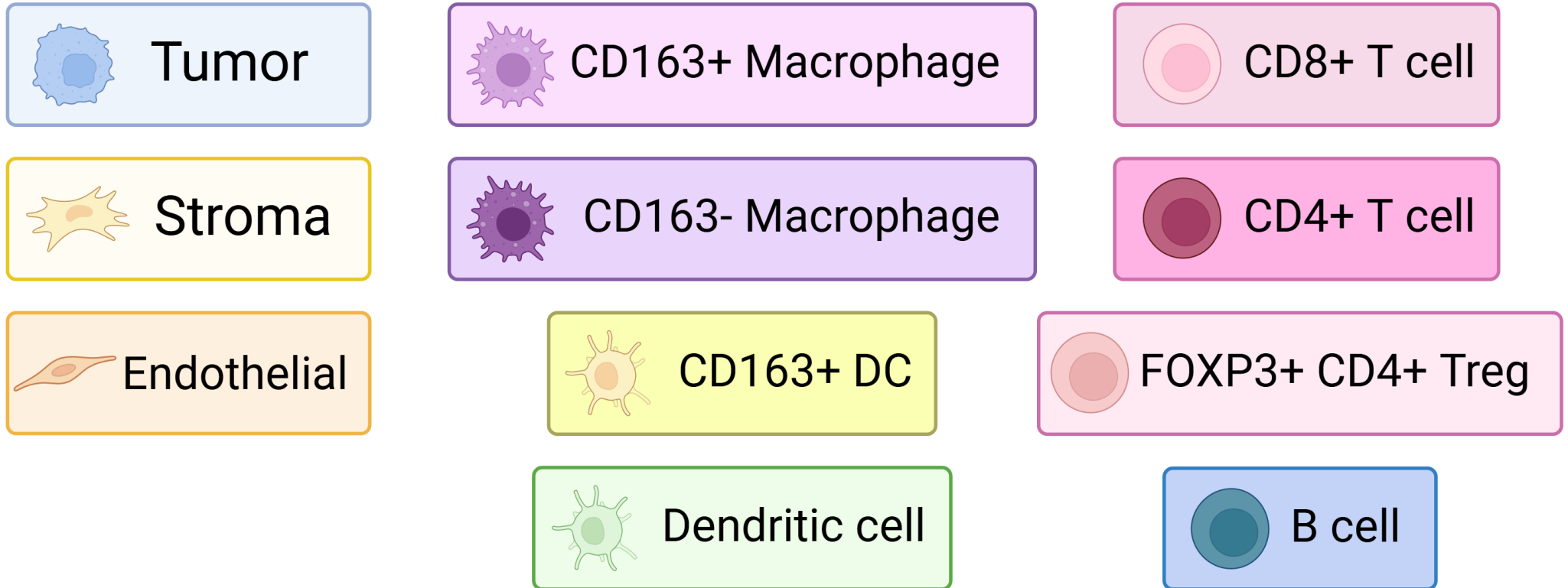
Phenotyping methods: Gating



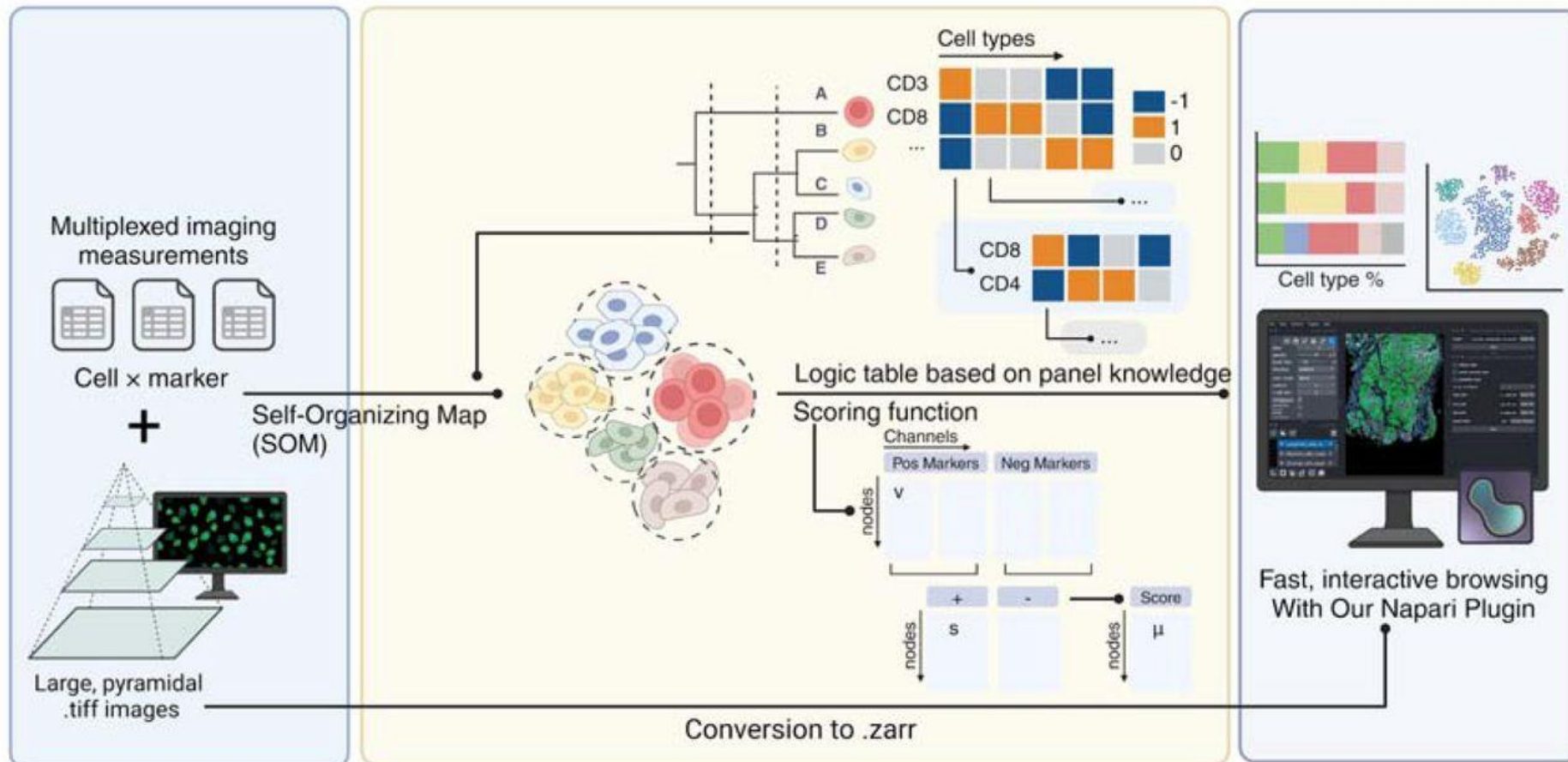
Phenotyping methods: TRIBUS

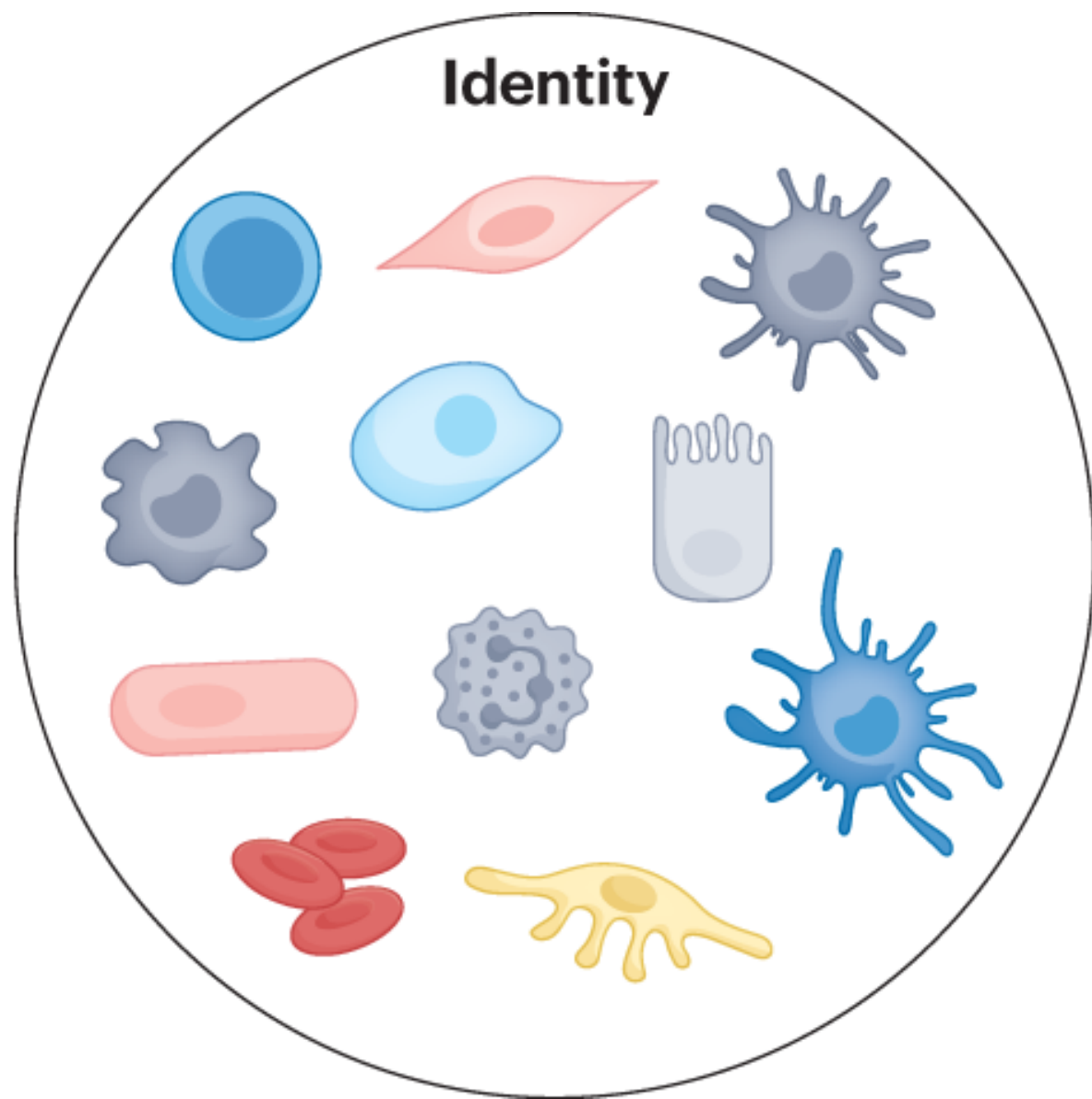


How can we assign phenotypes to our cells?

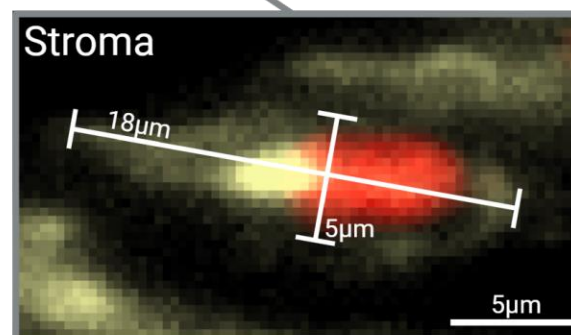
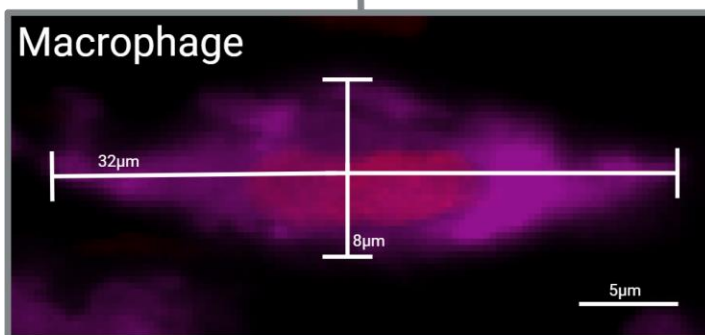
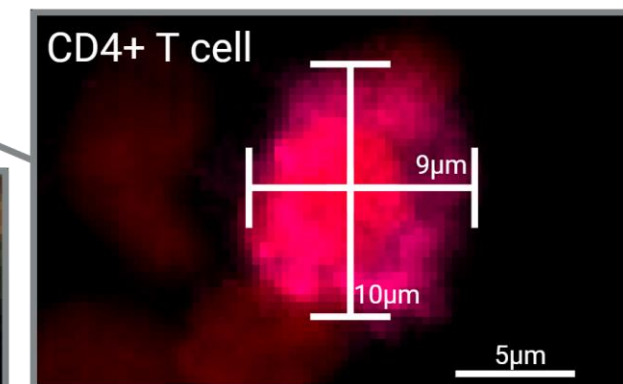
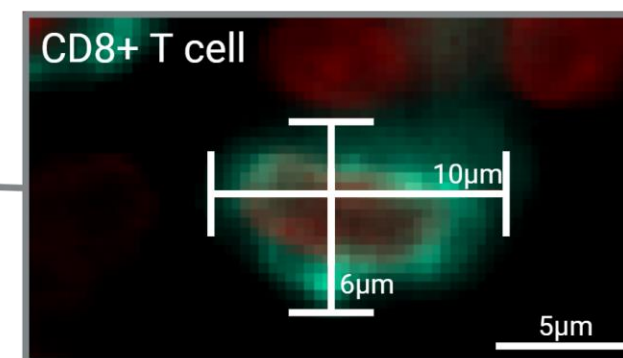
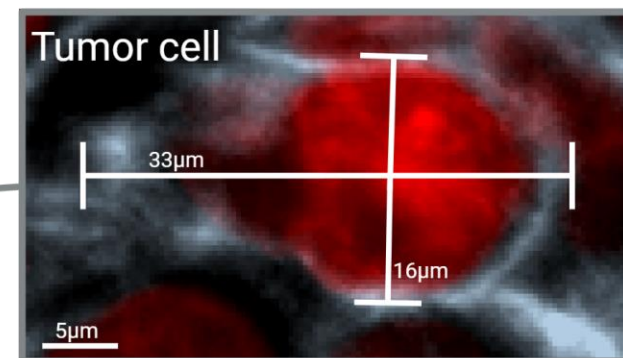
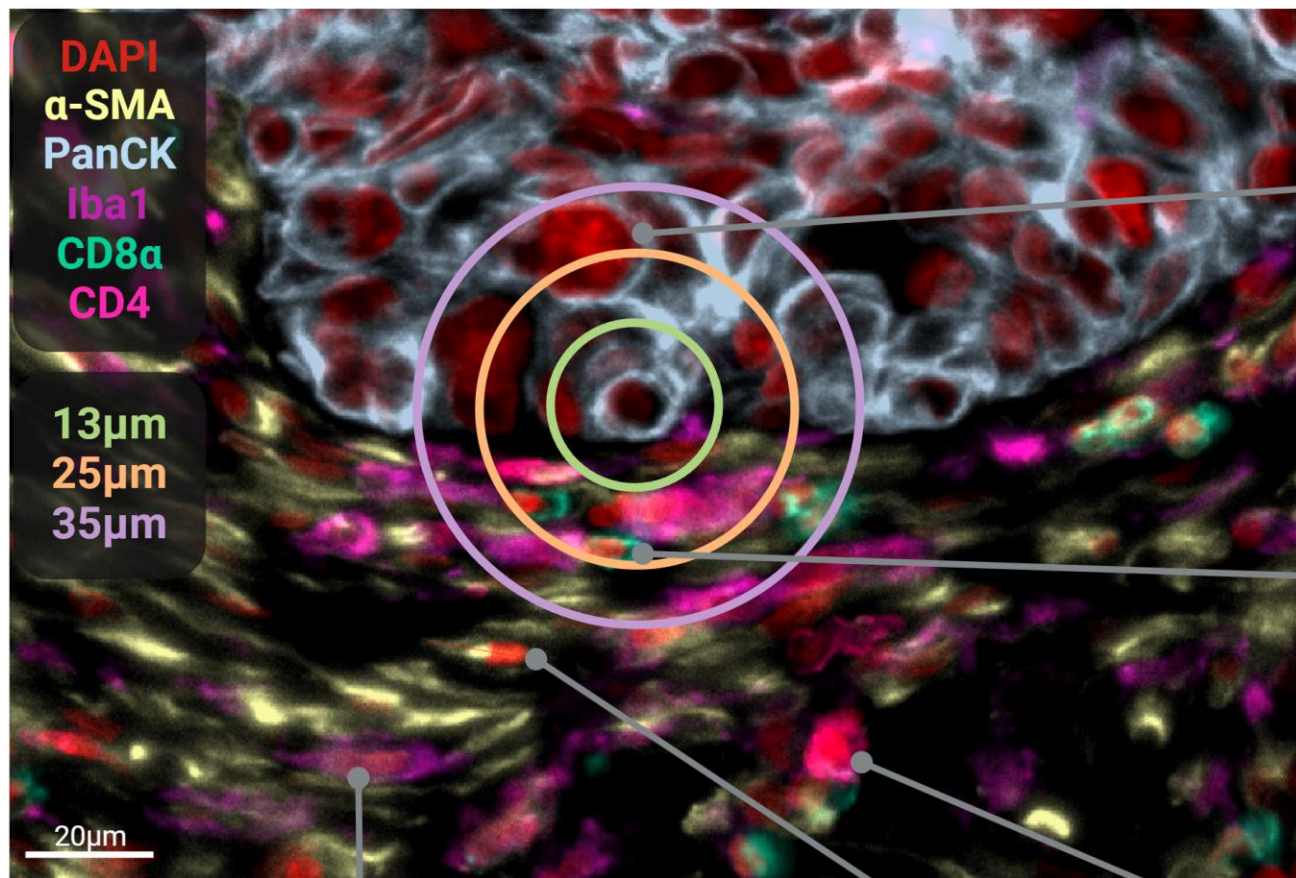


Phenotyping methods: TRIBUS



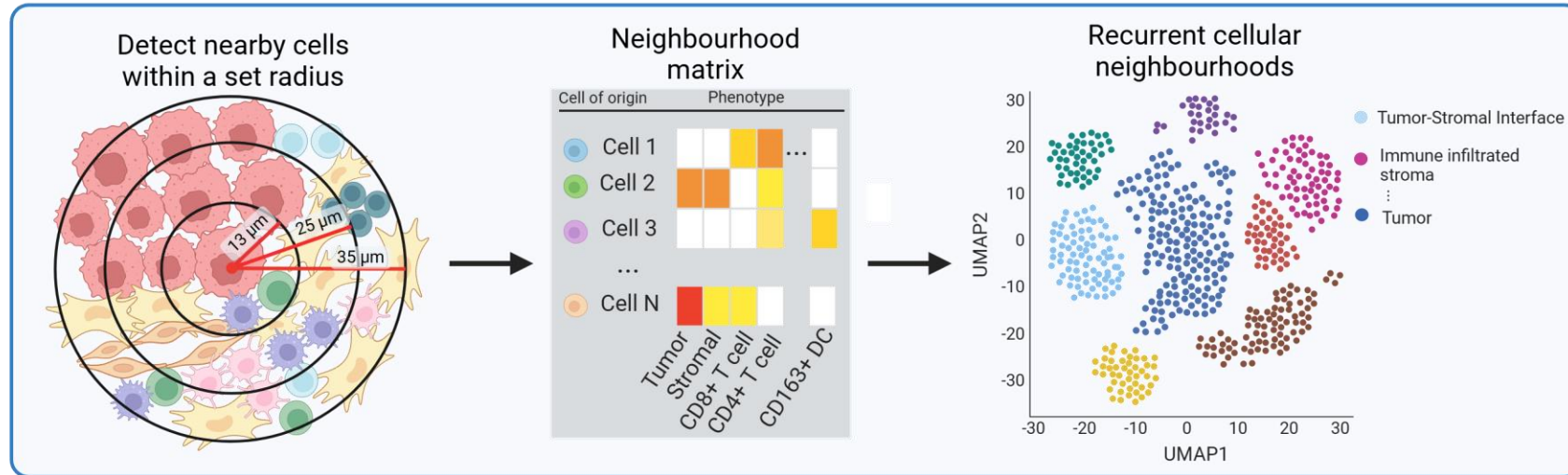


PORQUE EL TAMAÑO,
SÍ IMPORTA

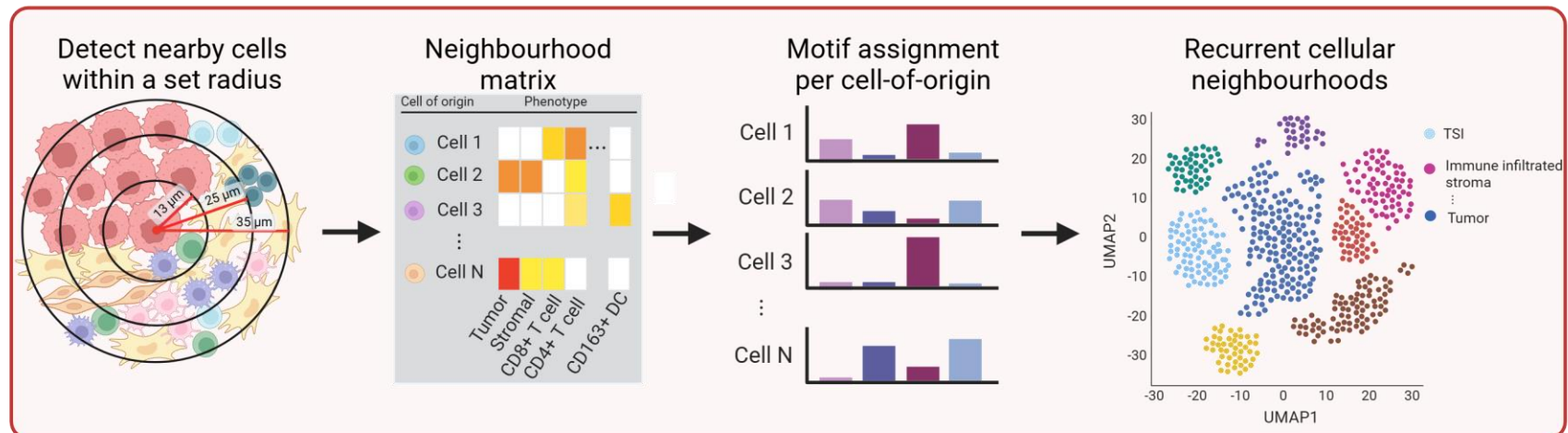


Understanding the distribution of cells in space

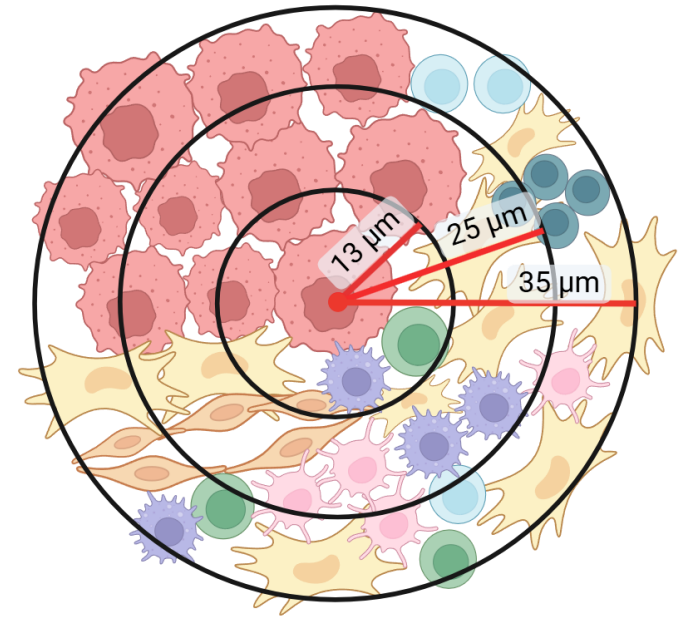
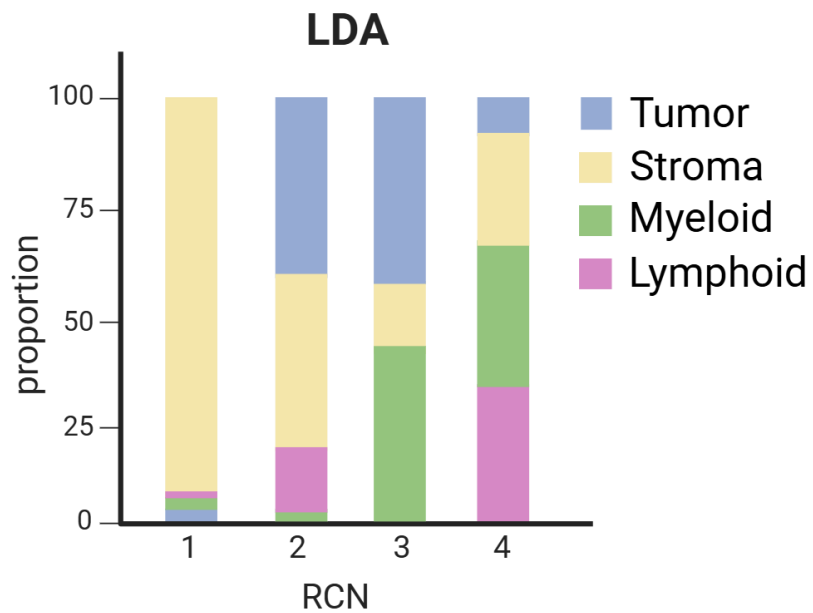
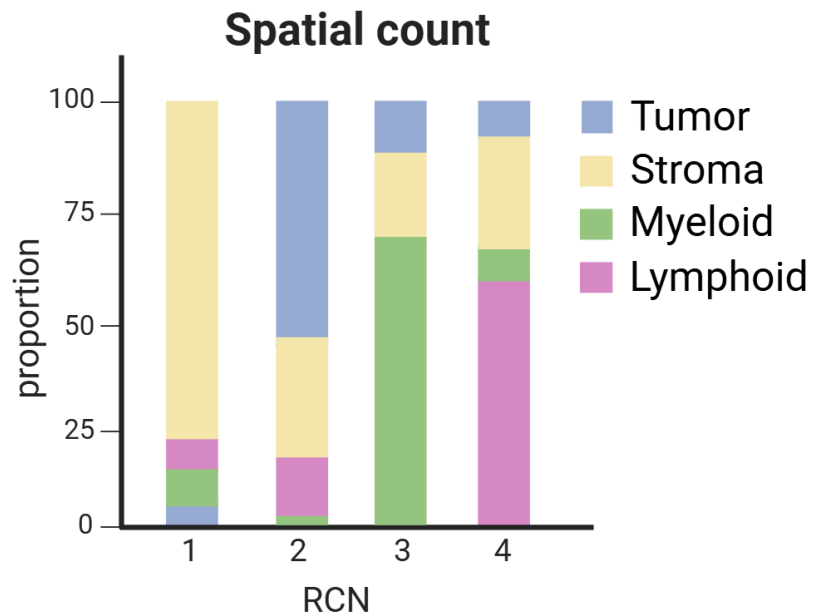
a. Workflow of spatial count

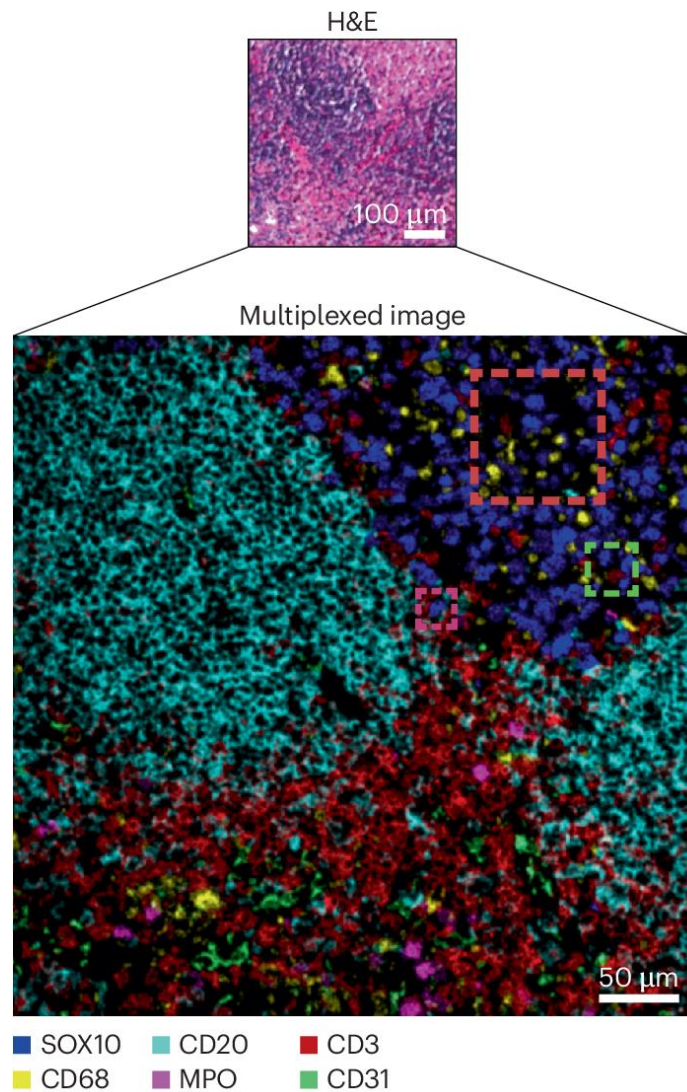


b. Workflow of LDA



Understanding the distribution of cells in space





Information at various length scales

